A 1.5D Field-Aligned Two-Fluid MHD Model of the Solar Chromosphere

Keheng Zhu

May 21, 2026

Abstract

We describe a 1.5D field-aligned two-fluid (ion + neutral) magnetohydrodynamic model for the solar chromosphere, following the formulation of Al Shidi et al. (2019). The general 3D two-fluid system is presented together with its multi-dimensional numerical framework (flux Jacobians, eigenvalues, TVD–MUSCL reconstruction, and semi-implicit time integration), and then reduced to the field-aligned form actually solved by the code. Quasi-neutrality is used to absorb electron pressure and electron heat conduction into the ion equations, giving a 6-variable system per field line. Hydrogen ionization and radiative recombination are included as point-source exchanges of mass, momentum, and energy between the two fluids, with explicit accounting for the 13.6 eV ionization-potential drain on the electron thermal pool, following Meier and Shumlak (2012) and Khomenko et al. (2014). Closure relations for electron and neutral heat conductivities, and for the ionization and recombination rate coefficients (Voronov, 1997; Hummer, 1994), are derived and validated against laboratory hydrogen data and the FAL Model C7 chromospheric profile (Avrett and Loeser, 2008).

1 Introduction

The solar chromosphere is a partially ionized layer in which ion–neutral interactions, gravitational stratification, field-aligned heat conduction, and hydrogen ionization/recombination all play leading-order roles. We solve a two-fluid (ion and neutral) MHD system reduced to flow along a prescribed magnetic field line, with electron contributions folded into the ion fluid via quasi-neutrality ($n_\mathrm{e} = n_\mathrm{i}$, $T_\mathrm{e} = T_\mathrm{i}$). Hydrogen ionization and radiative recombination are added as point-source exchanges between the two fluids, following the canonical reacting plasma–neutral framework of Meier and Shumlak (2012) under the electron-folded reduction of Khomenko et al. (2014). Section 2 gives the underlying 3D normalized equations, including the ionization/recombination source terms. Section 3 carries out the field-aligned reduction and assembles the 6-equation conservative system, defining the explicit residual $\mathbf{R}_{E,i}$ and the implicit residual $\mathbf{R}_{I,i}$ (which includes the ionization/recombination block). Section 4 specifies the numerical scheme: TVD–MUSCL reconstruction, the predictor–corrector that recovers second-order accuracy in time, and a five-stage semi-implicit time integrator (drag, temperature-equilibration, conduction, and ionization-recombination) that handles the stiff blocks of $\mathbf{R}_{I,i}$ by direct backward-Euler solves. Section 5 presents the heat-conductivity and ionization-rate closures and their validation. Section 6 shows initial numerical results with Model C7 as initial condition. Appendix A collects the general 3D multi-dimensional state vector, flux Jacobians, and eigenvalues; the ionization sources are pure body sources and do not modify the eigen system. This material provides background for the field-aligned reduction but is not used directly by the 1.5D implementation.

2 Two-Fluid MHD Model

2.1 Normalized Equations

$$\frac{\partial n_\mathrm{i}}{\partial t} + \nabla \cdot (n_\mathrm{i}\mathbf{V}) = \Gamma_\mathrm{net} \tag{1}$$

$$\frac{\partial n_\mathrm{n}}{\partial t} + \nabla \cdot (n_\mathrm{n}\mathbf{U}) = -\Gamma_\mathrm{net} \tag{2}$$

$$\frac{\partial n_{\rm i}\mathbf{V}}{\partial t} = -\nabla\cdot\left[n_{\rm i}\mathbf{V}\mathbf{V} + n_{\rm i}T_{\rm i} + \frac{B^2}{2}\mathbf{I} - \mathbf{B}\mathbf{B}\right] + n_{\rm i}\mathbf{g} - n_{\rm i}\nu_{\rm in}(\mathbf{V}-\mathbf{U}) + \Gamma_{\rm ion}\mathbf{U} - \Gamma_{\rm rec}\mathbf{V} \tag{3}$$

$$\frac{\partial n_{\rm n}\mathbf{U}}{\partial t} = -\nabla\cdot\left[n_{\rm n}\mathbf{U}\mathbf{U} + \frac{n_{\rm n}}{m_{\rm n}}T_{\rm n}\right] + n_{\rm n}\mathbf{g} + \frac{n_{\rm i}\nu_{\rm in}}{m_{\rm n}}(\mathbf{V}-\mathbf{U}) - \frac{1}{m_{\rm n}}(\Gamma_{\rm ion}\mathbf{U} - \Gamma_{\rm rec}\mathbf{V}) \tag{4}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla\cdot(\mathbf{V}\mathbf{B} - \mathbf{B}\mathbf{V}) - \nabla\times\left(\eta\mathbf{J} + \frac{1}{n_{\rm e}e}\mathbf{J}\times\mathbf{B}\right) \tag{5}$$

$$\begin{aligned}\frac{\partial T_{\rm i}}{\partial t} =& -\nabla\cdot(T_{\rm i}\mathbf{V}) + \frac{1}{3}T_{\rm i}\nabla\cdot\mathbf{V} - \frac{2}{3}\frac{\nabla\cdot(\mathbf{q}_{\rm i} + \eta\mathbf{B}\times\mathbf{J})}{n_{\rm i}} \\ &+ \frac{2}{3}\cdot\frac{1}{1+m_{\rm n}}\nu_{\rm in}\left[3\left(T_{\rm n} - T_{\rm i}\right) + m_{\rm n}|\mathbf{V}-\mathbf{U}|^2\right],\end{aligned} \tag{6}$$

$$\begin{aligned}\frac{\partial T_{\rm n}}{\partial t} =& -\nabla\cdot(T_{\rm n}\mathbf{U}) + \frac{1}{3}T_{\rm n}\nabla\cdot\mathbf{U} - \frac{2}{3}\frac{\nabla\cdot\mathbf{q}_{\rm n}}{n_{\rm n}} \\ &+ \frac{2}{3}\cdot\frac{m_{\rm n}}{1+m_{\rm n}}\nu_{\rm in}\left[3\left(T_{\rm i} - T_{\rm n}\right) + |\mathbf{V}-\mathbf{U}|^2\right],\end{aligned} \tag{7}$$

$$\nabla\times\mathbf{B} = \mathbf{J} \tag{8}$$

$$\mathbf{q} = -\kappa\nabla T. \tag{9}$$

The collision notation above should be read in terms of one momentum-transfer coefficient. In dimensional variables $\alpha = \rho_i\nu_{in} = \rho_n\nu_{ni}$; in the normalized variables of this subsection, with $m_i = 1$, $\alpha = n_i\nu_{in} = m_n n_n \nu_{ni}$. Thus the neutral momentum source written as $n_i\nu_{in}(\mathbf{V}-\mathbf{U})/m_n$ is the neutral acceleration produced by the same equal-and-opposite drag force.

2.2 Hyperbolic Divergence Cleaning (GLM)

To control numerical $\nabla\cdot\mathbf{B}$ errors, we introduce the generalized Lagrange multiplier Φ (Dedner et al., 2002). Equation (5) becomes

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla\cdot(\mathbf{V}\mathbf{B} - \mathbf{B}\mathbf{V}) - \nabla\times\left(\eta\mathbf{J} + \frac{1}{n_{\rm e}{\rm e}}\mathbf{J}\times\mathbf{B}\right) + \nabla\Phi, \tag{10}$$

with

$$\frac{\partial\Phi}{\partial t} + c_{\rm h}^2\nabla\cdot\boldsymbol{B} = -\frac{c_{\rm h}^2}{c_{\rm p}^2}\Phi. \tag{11}$$

2.3 Ionization–Recombination Source Terms

Hydrogen ionization and recombination exchange mass, momentum, and energy between the ion and neutral fluids. We follow the canonical three-fluid (electron, ion, neutral) reacting-mixture framework of Meier and Shumlak (2012). Under the same quasi-neutrality reduction $n_e = n_i$, $T_e = T_i$ used in §3, that framework reduces to two-fluid sources of the form given by Khomenko et al. (2014, Eqs. 32–36) and implemented in Godunov-type two-fluid chromospheric codes by Popescu Braileanu et al. (2021); Popescu Braileanu and Keppens (2022), Maneva et al. (2017), and Leake et al. (2012). The ionization-potential energy sink we adopt below matches the optically-thin analytical treatment of Snow and Hillier (2020).

Define the collisional ionization rate, the photoionization rate, the radiative recombination rate, and the net ionization rate,

$$\Gamma_{\rm coll} \equiv n_e n_n S_i(T_e), \qquad \Gamma_{\rm phot} \equiv n_n P_{\rm phot}, \qquad \Gamma_{\rm rec} \equiv n_e^2\alpha_r(T_e), \tag{12}$$

$$\Gamma_{\rm ion} \equiv \Gamma_{\rm coll} + \Gamma_{\rm phot}, \qquad \Gamma_{\rm net} \equiv \Gamma_{\rm ion} - \Gamma_{\rm rec}, \tag{13}$$

where $S_i(T_e)$ [m^3/s] is the electron-impact ionization rate coefficient, $\alpha_r(T_e)$ [m^3/s] is the case-B radiative recombination rate coefficient, and $P_{\rm phot}$ [s^{-1}] is the effective ground-state photoionization rate (per neutral atom). Numerical fits for S_i and α_r (Voronov, 1997; Hummer, 1994) are given in

§5.5; the $P_{\rm phot}$ closure is discussed below the rate definitions. The hydrogen ionization potential is $\chi_H = 13.6$ eV $= 2.179872 \times 10^{-18}$ J.

Carlsson and Stein (2002) show that the dominant chromospheric hydrogen ionization channel is *not* direct Lyman-continuum absorption from the ground state, but rather collisional excitation from $n = 1$ to $n = 2$ in the Lyα transition followed by photoionization of the $n = 2$ population by Balmer-continuum photons ($n = 2 \to$ continuum). For the two-state (ground + continuum) reduction adopted here, $P_{\rm phot}$ stands in for the effective rate of this two-step pathway. Carlsson and Stein (2002) report a chromospheric ionization/recombination timescale of 10^3–10^5 s; Leenaarts et al. (2007) adopt the corresponding "fixed radiative rates" approach in their 2D non-equilibrium ionization simulations. We adopt the geometric mean $P_{\rm phot} \sim 10^{-4}$ s^{-1} as the default single-rate closure; a height-dependent profile in the spirit of Carlsson and Leenaarts (2012) would replace this constant with $P_{\rm phot}(s)$ without changing the equations below.

Mass. The continuity equations gain the source $\pm\Gamma_{\rm net}$ shown in Eqs. (1)–(2). Mass is exchanged 1-for-1: $\partial_t(n_i + n_n) + \nabla \cdot (n_i\mathbf{V} + n_n\mathbf{U}) = 0$ still holds.

Momentum. New ions inherit the velocity $\mathbf{U}$ of the neutral from which they were ionized; new neutrals inherit the velocity $\mathbf{V}$ of the recombined ion. The corresponding sources are shown in Eqs. (3)–(4); they conserve total momentum $n_i\mathbf{V} + n_n\mathbf{U}$ by construction.

Energy. The energy sources are most cleanly written in the species total-energy form $e_s = \frac{3}{2}n_s k_B T_s + \frac{1}{2}\rho_s|\mathbf{v}_s|^2$ used later in §3. After folding electrons into the ion fluid under quasi-neutrality, the additions to the charged-fluid and neutral energy equations are

$$\boxed{\begin{aligned} Q^{e_i}_{\rm ion} &= +\Gamma_{\rm ion}\left(\tfrac{3}{2}k_B T_n + \tfrac{1}{2}m_n|\mathbf{U}|^2\right) - \Gamma_{\rm rec}\left(\tfrac{3}{2}k_B T_i + \tfrac{1}{2}m_i|\mathbf{V}|^2\right) - \Gamma_{\rm coll}\,\chi_H, \\ Q^{e_n}_{\rm ion} &= -\Gamma_{\rm ion}\left(\tfrac{3}{2}k_B T_n + \tfrac{1}{2}m_n|\mathbf{U}|^2\right) + \Gamma_{\rm rec}\left(\tfrac{3}{2}k_B T_i + \tfrac{1}{2}m_i|\mathbf{V}|^2\right). \end{aligned}} \tag{14}$$

The thermal and kinetic pieces are pure bookkeeping: every ionized neutral (collisional *or* photoionized) carries its thermal energy $\frac{3}{2}k_B T_n$ and kinetic energy $\frac{1}{2}m_n|\mathbf{U}|^2$ from the neutral pool to the charged-fluid pool, and vice versa for recombinations. The asymmetric $-\Gamma_{\rm coll}\chi_H$ term in $Q^{e_i}_{\rm ion}$ is the collisional ionization-potential drain: each *electron-impact* ionization removes 13.6 eV from the electron thermal pool to break the H bond. Photoionizations, by contrast, are powered by absorbed UV photons rather than by local kinetic temperature: the χ_H needed to break the bond comes from the radiation field, not from the fluid, so $\Gamma_{\rm phot}$ does *not* appear in the drain term (Carlsson and Stein, 2002). The recombination photon escapes the chromosphere as Lyman-continuum radiation under the optically-thin assumption (Carlsson and Stein, 2002; Leake et al., 2012; Snow and Hillier, 2020). Summing the two rows,

$$Q^{e_i}_{\rm ion} + Q^{e_n}_{\rm ion} = -\Gamma_{\rm coll}\,\chi_H, \tag{15}$$

so the total non-radiative fluid energy decreases at exactly the rate at which collisional ionizations occur; in steady state $\Gamma_{\rm ion} = \Gamma_{\rm rec}$, and the radiation field maintains the difference $\Gamma_{\rm phot}\chi_H$ that the plasma itself does not supply.

Consistency of α_r with the optically-thin assumption. Case-B is the appropriate recombination coefficient when ground-state-recombination photons escape; this matches the optically-thin χ_H drain above. Case-A would double-count ground-state recombinations as immediately re-ionized by the same Lyman-continuum photon, which is correct only for a fully optically-thick layer. We use case-B throughout; see Snow et al. (2023) for a more general two-fluid treatment with collisional and radiative ionization/recombination in a multi-level hydrogen model.

Kinetic vs. Saha equilibrium. The steady state of Eq. (13) with T_e fixed is the positive root of $n_{\rm tot}\alpha_r f^2 - (n_{\rm tot}S_i + P_{\rm phot})(1-f)f - P_{\rm phot}(1-f) = 0$. In the chromospheric limit where S_i is exponentially suppressed ($U = \chi_H/k_B T_e \gtrsim 20$) this reduces to a photoionization/recombination balance,

$$n_{\rm tot}\alpha_r f^2 + P_{\rm phot} f - P_{\rm phot} = 0 \implies f_{\rm eq} \approx \sqrt{\frac{P_{\rm phot}}{n_{\rm tot}\alpha_r}} \quad (\text{for } f \ll 1). \tag{16}$$

Without $P_{\rm phot}$, the chromospheric limit drives $f \to 0$ because $S_i(T_e \sim 6500\text{ K})$ is too small to balance recombination from any non-trivial f. The collisional kinetic fixed point $f/(1-f) = S_i/\alpha_r$ is recovered only in the upper chromosphere/transition region where T_e is high enough that $n_e S_i \gtrsim P_{\rm phot}$. This is the *kinetic* equilibrium and differs from the LTE Saha balance precisely because radiative recombination breaks detailed balance with photon escape. Leenaarts et al. (2007) showed that chromospheric ionization is generically out of LTE on time-dependent timescales much longer than the recombination time, so neither Saha nor any kinetic-equilibrium algebraic relation can be substituted for the time-dependent rate equations (13); see also the dynamic hydrogen ionization study of Carlsson and Stein (2002).

3 Field-Aligned Reduction

3.1 Field-Parallel Equations

We now assume the plasma flow is parallel or anti-parallel to the magnetic field, so $\mathbf{V} = V\mathbf{b}$ and $\mathbf{U} = U\mathbf{b}$. Substituting the T equations with energy equations, where $e = \frac{3}{2}p + \frac{1}{2}\rho v^2 + \rho\phi_g$ and $\nabla\phi_g = -\mathbf{g}$, we use one symmetric momentum-transfer coefficient

$$\alpha \equiv \rho_i \nu_{in} = \rho_n \nu_{ni}. \tag{17}$$

Here ν_{in} and ν_{ni} are reciprocal collision frequencies and are not generally equal. Writing the drag forces as $R_i = -\alpha(V - U)$ and $R_n = +\alpha(V - U)$ guarantees conservation of total momentum by ion–neutral collisions. We obtain

$$\frac{\partial \rho_{\rm i}}{\partial t} + \nabla \cdot (\rho_{\rm i} V \mathbf{b}) = m_{\rm i}\Gamma_{\rm net} \tag{18}$$

$$\frac{\partial \rho_{\rm n}}{\partial t} + \nabla \cdot (\rho_{\rm n} U \mathbf{b}) = -m_{\rm i}\Gamma_{\rm net} \tag{19}$$

$$\frac{\partial \rho_{\rm i} V}{\partial t} + \nabla \cdot \left[\rho_{\rm i} V^2 + n_{\rm i} k_{\rm b} T_{\rm i}\right] \mathbf{b} = (n_{\rm i} k_{\rm b} T_{\rm i}) \nabla \cdot \mathbf{b} - \rho_{\rm i} (\mathbf{b} \cdot \nabla)\phi_g - \alpha(V - U) + m_{\rm i}(\Gamma_{\rm ion} U - \Gamma_{\rm rec} V) \tag{20}$$

$$\frac{\partial \rho_{\rm n} U}{\partial t} + \nabla \cdot \left[\rho_{\rm n} U^2 + n_{\rm n} k_{\rm b} T_{\rm n}\right] \mathbf{b} = (n_{\rm n} k_{\rm b} T_{\rm n}) \nabla \cdot \mathbf{b} - \rho_{\rm n} (\mathbf{b} \cdot \nabla)\phi_g + \alpha(V - U) - m_{\rm i}(\Gamma_{\rm ion} U - \Gamma_{\rm rec} V) \tag{21}$$

$$\begin{aligned}\frac{\partial e_i}{\partial t} + \nabla \cdot [(e_i + n_i k_{\rm b} T_i) V \mathbf{b}] = &- \nabla \cdot \mathbf{q}_{\rm i} + \frac{\alpha}{m_{\rm i} + m_{\rm n}}[3k_{\rm b}(T_n - T_i) + m_{\rm n}|V - U|^2] - \alpha V(V - U) \\ &+ Q^{e_i}_{\rm ion},\end{aligned} \tag{22}$$

$$\begin{aligned}\frac{\partial e_n}{\partial t} + \nabla \cdot [(e_n + n_n k_{\rm b} T_n) U \mathbf{b}] = &- \nabla \cdot \mathbf{q}_{\rm n} + \frac{\alpha}{m_{\rm i} + m_{\rm n}}[3k_{\rm b}(T_i - T_n) + m_{\rm i}|V - U|^2] + \alpha U(V - U) \\ &+ Q^{e_n}_{\rm ion},\end{aligned} \tag{23}$$

with $Q^{e_i}_{\rm ion}$ and $Q^{e_n}_{\rm ion}$ defined in Eq. (14) (specialized to the field-aligned scalars V, U, with $|\mathbf{V}|^2 \to V^2$ and $|\mathbf{U}|^2 \to U^2$).

$$\mathbf{q} = -\kappa \nabla T, \tag{24}$$

where the form of $\mathbf{q}$ is uncertain. An alternative, suggested by Xianyu, is $\mathbf{q} = -\kappa \mathbf{b}\mathbf{b} \cdot \nabla T$. In the 1.5D field-aligned reduction, both $\mathbf{q} = -\kappa\nabla T$ and $\mathbf{q} = -\kappa\mathbf{b}, \mathbf{b} \cdot \nabla T$ reduce to $\mathbf{q} = -\kappa\mathbf{b}, \partial_s T$, and therefore $-\nabla \cdot \mathbf{q} = B\partial_s[(\kappa/B)\partial_s T]$. The distinction matters only in multidimensional models with perpendicular temperature gradients.

3.2 Folding in Electrons via Quasi-Neutrality

These equations have a clear limitation: electron pressure and electron heat conduction are neglected, despite the assumption $T_e = T_i$. Imposing quasi-neutrality $n_i = n_e$, electron effects are incorporated into the equation of motion and the energy equation via

$$p_i = n_{\rm i} k_{\rm b} T_{\rm i} + n_{\rm e} k_{\rm b} T_{\rm e} = 2 n_{\rm i} k_{\rm b} T_{\rm i}, \tag{25}$$

and

$$\mathbf{q_e} = -\kappa_{e,0} T_e^{5/2} \mathbf{b}\, \partial_s T_e, \quad \kappa_{e,0} \approx 9.2 \times 10^{-12} \text{ W m}^{-1} \text{ K}^{-7/2}. \tag{26}$$

Below, κ_e without the subscript "0" denotes the full electron thermal conductivity multiplying $\partial_s T_e$.

The equations become

$$\frac{\partial \rho_{\rm i}}{\partial t} + \nabla \cdot (\rho_{\rm i} V \mathbf{b}) = m_{\rm i}\Gamma_{\rm net} \tag{27}$$

$$\frac{\partial \rho_{\rm n}}{\partial t} + \nabla\cdot(\rho_{\rm n} U \mathbf{b}) = -m_{\rm i}\Gamma_{\rm net} \tag{28}$$

$$\frac{\partial \rho_{\rm i} V}{\partial t} + \nabla\cdot\left[\rho_{\rm i} V^2 + 2n_{\rm i}k_{\rm b}T_{\rm i}\right]\mathbf{b} = (2n_{\rm i}k_{\rm b}T_{\rm i})\nabla\cdot\mathbf{b} - \rho_{\rm i}(\mathbf{b}\cdot\nabla)\phi_g - \alpha(V-U) + m_{\rm i}(\Gamma_{\rm ion}U - \Gamma_{\rm rec}V) \tag{29}$$

$$\frac{\partial \rho_{\rm n} U}{\partial t} + \nabla\cdot\left[\rho_{\rm n} U^2 + n_{\rm n}k_{\rm b}T_{\rm n}\right]\mathbf{b} = (n_{\rm n}k_{\rm b}T_{\rm n})\nabla\cdot\mathbf{b} - \rho_{\rm n}(\mathbf{b}\cdot\nabla)\phi_g + \alpha(V-U) - m_{\rm i}(\Gamma_{\rm ion}U - \Gamma_{\rm rec}V) \tag{30}$$

$$\begin{aligned}\frac{\partial e_i}{\partial t} + \nabla\cdot\left[(e_i + 2n_ik_{\rm b}T_i)V\mathbf{b}\right] = &-\nabla\cdot\mathbf{q}_{\rm i} - \nabla\cdot\mathbf{q}_{\rm e} + \frac{\alpha}{m_{\rm i}+m_{\rm n}}\left[3k_{\rm b}(T_n - T_i) + m_{\rm n}|V-U|^2\right] - \alpha V(V-U)\\ &+ Q_{\rm ion}^{e_i},\end{aligned} \tag{31}$$

$$\begin{aligned}\frac{\partial e_n}{\partial t} + \nabla\cdot\left[(e_n + n_nk_{\rm b}T_n)U\mathbf{b}\right] = &-\nabla\cdot\mathbf{q}_{\rm n} + \frac{\alpha}{m_{\rm i}+m_{\rm n}}\left[3k_{\rm b}(T_i - T_n) + m_{\rm i}|V-U|^2\right] + \alpha U(V-U)\\ &+ Q_{\rm ion}^{e_n}.\end{aligned} \tag{32}$$

The χ_H drain in $Q_{\rm ion}^{e_i}$ acts on the charged-fluid energy, i.e. on the combined electron+ion thermal pool, exactly as intended by the quasi-neutrality reduction: ionization removes 13.6 eV per event from the electron thermal energy, which under the electron-folded convention is part of e_i.

Here ρ_i, V, and e_i denote quantities associated with charged particles, including both ions and electrons, while n_i and T_i are specific to ions. Because ions dominate the charged-particle mass and momentum, ρ_i, V, and e_i can be reasonably approximated as ion-related: for instance, $\rho_i \approx m_i n_i$, where m_i is the ion mass.

Because electrons move significantly faster than ions and neutrals, electron heat conduction dominates over ion and neutral heat conduction. In regions of low ionization, however, neutral heat conduction can become comparable or larger; this is treated quantitatively in Section 5.

3.3 Conservative Vector Form

The reduced equations can be rewritten in conservation form with derivatives along the field line s as

$$\frac{\partial}{\partial t}\begin{bmatrix}\rho_i\\ \rho_n\\ \rho_i V\\ \rho_n U\\ e_i\\ e_n\end{bmatrix} + B\frac{\partial}{\partial s}\begin{bmatrix}\frac{\rho_i V}{B}\\ \frac{\rho_n U}{B}\\ \frac{\rho_i V^2 + 2n_ik_bT_i}{B}\\ \frac{\rho_n U^2 + n_nk_bT_n}{B}\\ \frac{(e_i + 2n_ik_bT_i)V}{B}\\ \frac{(e_n + n_nk_bT_n)U}{B}\end{bmatrix} = \mathbf{S}, \tag{33}$$

where

$$\mathbf{S} = \begin{bmatrix} +m_i\Gamma_{\rm net}\\ -m_i\Gamma_{\rm net}\\ 2n_ik_bT_iB\frac{\partial}{\partial s}(\frac{1}{B}) - \rho_i\frac{\partial}{\partial s}\phi_g - \alpha(V-U) + m_i(\Gamma_{\rm ion}U - \Gamma_{\rm rec}V)\\ n_nk_bT_nB\frac{\partial}{\partial s}(\frac{1}{B}) - \rho_n\frac{\partial}{\partial s}\phi_g + \alpha(V-U) - m_i(\Gamma_{\rm ion}U - \Gamma_{\rm rec}V)\\ B\frac{\partial}{\partial s}\left[\frac{\kappa_i+\kappa_e}{B}\frac{\partial T_i}{\partial s}\right] + \frac{\alpha}{m_i+m_n}\left[3k_b(T_n-T_i) + m_n|V-U|^2\right] - \alpha V(V-U) + Q_{\rm ion}^{e_i}\\ B\frac{\partial}{\partial s}\left[\frac{\kappa_n}{B}\frac{\partial T_n}{\partial s}\right] + \frac{\alpha}{m_i+m_n}\left[3k_b(T_i-T_n) + m_i|V-U|^2\right] + \alpha U(V-U) + Q_{\rm ion}^{e_n} \end{bmatrix} \tag{34}$$

The six ionization additions are pure point sources, with no field-line derivative; the flux divergence on the left-hand side of the conservative form, and therefore the eigen system of §3.4, is unchanged.

Here the electron conductivity in the ion energy source is the full $\kappa_e(T_i, n_i, n_n)$, not the Spitzer prefactor $\kappa_{e,0}$. The use of α also makes the collisional energy balance explicit. If $w = V - U$, the drag forces do work on the two bulk kinetic energies at rate $VR_i + UR_n = -\alpha w^2$. The two collisional thermal terms sum to αw^2, so ion–neutral collisions exchange momentum and convert drift kinetic energy into heat without changing the total energy of the combined ion–neutral system.

The pressure-area term in the momentum sources is not an additional pressure force. For any scalar X, the prescribed field satisfies $\nabla\cdot(B\mathbf{b}) = 0$, hence

$$\nabla\cdot(X\mathbf{b}) = B\frac{\partial}{\partial s}\left(\frac{X}{B}\right). \tag{35}$$

For example,

$$B\frac{\partial}{\partial s}\left(\frac{\rho_i V^2 + p_i}{B}\right) - p_i B\frac{\partial}{\partial s}\left(\frac{1}{B}\right) = B\frac{\partial}{\partial s}\left(\frac{\rho_i V^2}{B}\right) + \frac{\partial p_i}{\partial s}, \tag{36}$$

where $p_i = 2n_i k_b T_i$. Thus the flux divergence and geometric source together recover the ordinary pressure-gradient force. In contrast, the heat conduction source must retain the complete field-line diffusion operator. Writing $K_i = \kappa_i + \kappa_e$, its expanded form is

$$B\frac{\partial}{\partial s}\left(\frac{K_i}{B}\frac{\partial T_i}{\partial s}\right) = K_i\frac{\partial^2 T_i}{\partial s^2} + \left[\frac{\partial K_i}{\partial s} + K_i B\frac{\partial}{\partial s}\left(\frac{1}{B}\right)\right]\frac{\partial T_i}{\partial s}, \tag{37}$$

with the same structure for κ_n and T_n. The second-derivative term is therefore part of the source and is handled by the diffusion solve.

Dividing through by B, this is equivalent to

$$\frac{\partial}{\partial t}\left(\frac{\mathbf{U}}{B}\right) + \frac{\partial}{\partial s}\left(\frac{\mathbf{F}}{B}\right) = \frac{\mathbf{S}}{B} \tag{38}$$

Equation (38) has the same form as the standard conservation law $\partial_t \mathbf{U}' + \partial_s \mathbf{F}' = \mathbf{S}'$, so the eigen system of the next subsection follows by the same construction as in Appendix A.

3.4 Eigen System

The eigenvalues are

$$\begin{pmatrix} V \\ U \\ V - \sqrt{2\gamma k_b n_i T_i/\rho_i} \\ V + \sqrt{2\gamma k_b n_i T_i/\rho_i} \\ U - \sqrt{\gamma k_b n_n T_n/\rho_n} \\ U + \sqrt{\gamma k_b n_n T_n/\rho_n} \end{pmatrix} \tag{39}$$

where $\gamma = \frac{5}{3}$. Without the induction equation, c_{fast} degenerates to the sound speed $c_s = \sqrt{\gamma p/\rho}$. The spectral radius of the Jacobian is therefore

$$\rho\left(\frac{\partial \mathbf{F}'}{\partial \mathbf{U}'}\right) = \max(|V| + |c_{s,i}|, |U| + |c_{s,n}|). \tag{40}$$

4 Numerical Scheme

We now specify the spatial and temporal discretizations applied to the field-aligned conservative form, Eq. (38), written per cell as

$$\frac{\partial \mathbf{U}_i}{\partial t} = \mathbf{R}_{E,i}(\mathbf{U}) + \mathbf{R}_{I,i}(\mathbf{U}) = -\frac{B_i}{ds_i}\left(\frac{\mathbf{F}_{i+\frac{1}{2}}}{B_{i+\frac{1}{2}}} - \frac{\mathbf{F}_{i-\frac{1}{2}}}{B_{i-\frac{1}{2}}}\right) + \mathbf{S}_{E,i} + \mathbf{R}_{I,i}. \tag{41}$$

Section 4.1 introduces the TVD–MUSCL reconstruction and Rusanov numerical flux for an abstract scalar conservation law; §4.2 specializes them to the field-aligned system and assembles the explicit and implicit residuals $\mathbf{R}_{E,i}$ and $\mathbf{R}_{I,i}$. A predictor–corrector step that recovers second-order accuracy in time is described in §4.3, and the four-stage semi-implicit time integration of $\mathbf{R}_{I,i}$ — which treats its three physically distinct blocks by direct backward-Euler solves — is laid out in §4.4.

4.1 TVD–MUSCL Reconstruction and Numerical Flux

We discretize the flux divergence in Eq. (41) with a TVD–MUSCL reconstruction in primitive variables followed by a Rusanov (local Lax–Friedrichs) numerical flux. The construction here applies to the abstract scalar conservation law $\partial_t u + \partial_x f(u) = 0$; it specializes to the field-aligned system by reading $u \to \mathbf{W}$, $f \to \mathbf{F}(\mathbf{W})$ as defined in Eq. (47).

A simple upwind scheme for the nonlinear flux is

$$\hat{F}_{j+1/2} = \frac{1}{2}\left(f_j + f_{j+1}\right) - \frac{1}{2}\left|\hat{a}_{j+1/2}\right|\left(u_{j+1} - u_j\right), \tag{42}$$

where the wave speed $\hat{a}_{j+1/2}$ is defined as

$$\hat{a}_{j+1/2} = \begin{cases} \frac{f_{j+1}-f_j}{u_{j+1}-u_j} & \text{when } u_{j+1} \neq u_j, \\ f'\left(u_j\right) & \text{when } u_{j+1} = u_j. \end{cases} \tag{43}$$

For higher-order spatial accuracy, the actual solver uses a TVD–MUSCL reconstruction. The numerical flux at $j \pm 1/2$ is taken in Rusanov / local Lax–Friedrichs form,

$$\begin{aligned} f^*_{j-1/2} &= \frac{1}{2}\left(\left[f\left(u^{\mathrm{r}}_{j-1/2}\right) + f\left(u^{\mathrm{l}}_{j-1/2}\right)\right] - \lambda_{j-1/2}\left[u^{\mathrm{r}}_{j-1/2} - u^{\mathrm{l}}_{j-1/2}\right]\right), \\ f^*_{j+1/2} &= \frac{1}{2}\left(\left[f\left(u^{\mathrm{r}}_{j+1/2}\right) + f\left(u^{\mathrm{l}}_{j+1/2}\right)\right] - \lambda_{j+1/2}\left[u^{\mathrm{r}}_{j+1/2} - u^{\mathrm{l}}_{j+1/2}\right]\right), \end{aligned} \tag{44}$$

where $\lambda_j = \max\{\rho[\frac{\partial f}{\partial u}\left(u^l_j\right)], \rho[\frac{\partial f}{\partial u}\left(u^r_j\right)]\}$ and $\rho[A]$ denotes the spectral radius of matrix A, and where

$$\begin{aligned} u^{\mathrm{l}}_{j-1/2} &= u_{j-1} + \frac{1}{2}\phi\left(r_{j-1}\right)\left(u_j - u_{j-1}\right), \\ u^{\mathrm{l}}_{j+1/2} &= u_j + \frac{1}{2}\phi\left(r_j\right)\left(u_{j+1} - u_j\right), \\ u^{\mathrm{r}}_{j-1/2} &= u_j - \frac{1}{2}\phi\left(r_j\right)\left(u_{j+1} - u_j\right), \\ u^{\mathrm{r}}_{j+1/2} &= u_{j+1} - \frac{1}{2}\phi\left(r_{j+1}\right)\left(u_{j+2} - u_{j+1}\right), \\ r_j &= \frac{u_j - u_{j-1}}{u_{j+1} - u_j}, \\ \phi(r) &= \max(0, \min(1, r)), \lim_{r\to\infty} \phi(r) = 1. \end{aligned} \tag{45}$$

Specialized to the field-aligned system, these are precisely the interface states $\mathbf{W}^{L,R}_{i\pm1/2}$ and the Rusanov flux $\mathbf{F}^*_{i\pm1/2}$ of Eqs. (49).

4.2 Discrete Residuals on the Field-Aligned Grid

We now specialize the abstract TVD–MUSCL reconstruction and Rusanov flux of §4.1 to the field-aligned two-fluid system, and assemble the explicit and implicit residuals $\mathbf{R}_{E,i}$ and $\mathbf{R}_{I,i}$, together with the conduction operator $\mathcal{C}_{s,i}$, on the field-aligned grid. Let

$$\mathbf{W} = \begin{pmatrix} \rho_i & \rho_n & V & U & T_i & T_n \end{pmatrix}^T \tag{46}$$

be the primitive state reconstructed by the limiter. The conservative state and physical flux evaluated from $\mathbf{W}$ are

$$\mathbf{U}(\mathbf{W}) = \begin{pmatrix} \rho_i \\ \rho_n \\ \rho_i V \\ \rho_n U \\ \frac{3}{2}p_i + \frac{1}{2}\rho_i V^2 + \rho_i \phi_g \\ \frac{3}{2}p_n + \frac{1}{2}\rho_n U^2 + \rho_n \phi_g \end{pmatrix}, \quad \mathbf{F}(\mathbf{W}) = \begin{pmatrix} \rho_i V \\ \rho_n U \\ \rho_i V^2 + p_i \\ \rho_n U^2 + p_n \\ (e_i + p_i)V \\ (e_n + p_n)U \end{pmatrix}, \tag{47}$$

where $p_i = 2n_i k_b T_i$ and $p_n = n_n k_b T_n$.

For each cell interface $i + 1/2$, the left and right primitive states $\mathbf{W}^L_{i+1/2}$ and $\mathbf{W}^R_{i+1/2}$ are reconstructed componentwise with the minmod limiter (§4.1–§4.3). The corresponding conservative states are

$$\mathbf{U}^L_{i+1/2} = \mathbf{U}(\mathbf{W}^L_{i+1/2}), \qquad \mathbf{U}^R_{i+1/2} = \mathbf{U}(\mathbf{W}^R_{i+1/2}). \tag{48}$$

The Rusanov flux at the interface is then

$$\mathbf{F}^*_{i+1/2} = \frac{1}{2}\left[\mathbf{F}(\mathbf{W}^L_{i+1/2}) + \mathbf{F}(\mathbf{W}^R_{i+1/2}) - a_{i+1/2}\left(\mathbf{U}^R_{i+1/2} - \mathbf{U}^L_{i+1/2}\right)\right], \tag{49}$$

with

$$a_{i+1/2} = \max\left(|V^L| + c^L_{s,i}, |U^L| + c^L_{s,n}, |V^R| + c^R_{s,i}, |U^R| + c^R_{s,n}\right), \tag{50}$$

where $c_{s,i} = \sqrt{\gamma p_i/\rho_i}$ and $c_{s,n} = \sqrt{\gamma p_n/\rho_n}$.

Substituting the interface fluxes into Eq. (38) gives the explicit residual for cell i:

$$\mathbf{R}_{E,i} = -\frac{B_i}{\Delta s_i}\left(\frac{\mathbf{F}^*_{i+1/2}}{B_{i+1/2}} - \frac{\mathbf{F}^*_{i-1/2}}{B_{i-1/2}}\right) + \mathbf{S}_{E,i}. \tag{51}$$

For the split used here, the explicit source contains the pressure-area and gravitational terms,

$$\mathbf{S}_{E,i} = \begin{pmatrix} 0 \\ 0 \\ p_{\mathrm{i},i}B_iD_i - \rho_{\mathrm{i},i}G_i \\ p_{\mathrm{n},i}B_iD_i - \rho_{\mathrm{n},i}G_i \\ 0 \\ 0 \end{pmatrix}, \tag{52}$$

where

$$p_{\mathrm{i},i} = 2n_{\mathrm{i},i}k_bT_{\mathrm{i},i}, \qquad p_{\mathrm{n},i} = n_{\mathrm{n},i}k_bT_{\mathrm{n},i}, \qquad D_i = \frac{\partial}{\partial s}\left(\frac{1}{B}\right)\bigg|_i, \qquad G_i = \frac{\partial\phi_g}{\partial s}\bigg|_i. \tag{53}$$

The implicit residual contains the ion–neutral drag, collisional heating, heat conduction, and the ionization/recombination source rows from §2.3. In the same component order as $\mathbf{U}$,

$$\mathbf{R}_{I,i} = \begin{pmatrix} +m_i\Gamma_{\mathrm{net},i} \\ -m_i\Gamma_{\mathrm{net},i} \\ -\alpha_iw_i + m_i(\Gamma_{\mathrm{ion},i}U_i - \Gamma_{\mathrm{rec},i}V_i) \\ +\alpha_iw_i - m_i(\Gamma_{\mathrm{ion},i}U_i - \Gamma_{\mathrm{rec},i}V_i) \\ \mathcal{C}_{\mathrm{i},i} + \dfrac{\alpha_i}{m_i+m_n}\left[3k_b(T_{\mathrm{n},i} - T_{\mathrm{i},i}) + m_nw_i^2\right] - \alpha_iV_iw_i + Q^{e_i}_{\mathrm{ion},i} \\ \mathcal{C}_{\mathrm{n},i} + \dfrac{\alpha_i}{m_i+m_n}\left[3k_b(T_{\mathrm{i},i} - T_{\mathrm{n},i}) + m_iw_i^2\right] + \alpha_iU_iw_i + Q^{e_n}_{\mathrm{ion},i} \end{pmatrix}, \tag{54}$$

where $w_i = V_i - U_i$ and $\alpha_i = \rho_{\mathrm{i},i}\nu_{in,i} = \rho_{\mathrm{n},i}\nu_{ni,i}$. The ionization terms belong in $\mathbf{R}_I$ rather than $\mathbf{R}_E$ because the effective relaxation rate $\lambda_{\mathrm{ion}} \equiv n_e(S_i + \alpha_r)$ can satisfy $\lambda_{\mathrm{ion}}\Delta t \gg 1$ at the acoustic CFL in the transition region (see the discussion in §4.4, "Why split into explicit and implicit residuals" and "Why not Picard"). The residual is not evaluated as a single fixed point of $\mathbf{R}_{I,i}(\mathbf{U}^{n+1})$; instead, it is split into a drag block, a temperature-equilibration block, a heat-conduction block, and an ionization-recombination block, each solved by its own backward-Euler stage as described in Section 4.4.

The field-aligned heat conduction terms are discretized in conservative flux form:

$$\mathcal{C}_{\mathrm{i},i} = \frac{B_i}{\Delta s_i}\left[\frac{K_{\mathrm{i},i+1/2}}{B_{i+1/2}}\frac{T_{\mathrm{i},i+1} - T_{\mathrm{i},i}}{\Delta s_{i+1/2}} - \frac{K_{\mathrm{i},i-1/2}}{B_{i-1/2}}\frac{T_{\mathrm{i},i} - T_{\mathrm{i},i-1}}{\Delta s_{i-1/2}}\right], \tag{55}$$

where $K_{\mathrm{i}} = \kappa_i + \kappa_e(T_i, n_i, n_n)$ for the charged-fluid energy equation and $K_{\mathrm{i},i\pm1/2}$ denotes the face value. The neutral conduction term is

$$\mathcal{C}_{\mathrm{n},i} = \frac{B_i}{\Delta s_i}\left[\frac{\kappa_{\mathrm{n},i+1/2}}{B_{i+1/2}}\frac{T_{\mathrm{n},i+1} - T_{\mathrm{n},i}}{\Delta s_{i+1/2}} - \frac{\kappa_{\mathrm{n},i-1/2}}{B_{i-1/2}}\frac{T_{\mathrm{n},i} - T_{\mathrm{n},i-1}}{\Delta s_{i-1/2}}\right]. \tag{56}$$

With these residuals, the semi-implicit update used for the conservative vector form is

$$\mathbf{U}^{n+1}_i = \mathbf{U}^n_i + \Delta t_i\left[\mathbf{R}_{E,i}(\mathbf{U}^n) + \mathbf{R}_{I,i}(\mathbf{U}^{n+1})\right]. \tag{57}$$

The implicit residual is realized by the four-stage operator split of Section 4.4 (Eqs. (67)–(70)): an explicit MUSCL+Rusanov update of R_E, followed by an analytic drag/frictional-heating solve (Eqs. (82)–(85)), an analytic temperature-equilibration solve (Eqs. (91)–(92)), and a tridiagonal conduction solve (Eqs. (97)–(98)). The MUSCL prediction step uses the same B-weighted flux difference as Eq. (51).

4.3 Second-Order Accuracy in Time

The single-stage reconstruction above is first-order in time. To match the spatial order, the actual update couples the limiter to a predictor at $t^n + \Delta t/2$. Let U denote conservative variables and u primitive variables; $f(u)$ and $f(U)$ both stand for the flux function, which can be evaluated either way. We first compute the predictor $\tilde{U}^{n+1} = U^n - \frac{\Delta t}{\Delta x}[f(u^l_{i+\frac{1}{2}}) - f(u^r_{i-\frac{1}{2}})]$ along with $\tilde{u}^{n+1} = u(\tilde{U}^{n+1})$. The 2nd-order reconstruction is then

$$\begin{aligned} u^L_{i+\frac{1}{2}} &= \frac{1}{2}(u^n_i + \tilde{u}^{n+1}_i) + \phi(r_i)(u^n_{i+1} - u^n_i) \\ u^R_{i+\frac{1}{2}} &= \frac{1}{2}(u^n_{i+1} + \tilde{u}^{n+1}_{i+1}) - \phi(r_{i+1})(u^n_{i+2} - u^n_{i+1}), \end{aligned} \tag{58}$$

and the numerical fluxes are

$$F_{i+\frac{1}{2}} = \frac{1}{2}([f(u^L_{i+\frac{1}{2}}) + f(u^R_{i+\frac{1}{2}})] - \lambda_{i+1/2}[U(u^R_{i+\frac{1}{2}}) - U(u^L_{i+\frac{1}{2}})]). \tag{59}$$

In this way, even though we use forward Euler time stepping (next subsection), the overall scheme is still second-order accurate in time. Primitive variables are used for reconstruction to suppress their errors.

4.4 Semi-implicit Time Integration

With $\mathbf{R}_{E,i}$ now fully specified by §4.1–§4.3, we turn to time integration of Eq. (41). A semi-implicit Euler time stepping is adopted, because the generalized Lagrange multiplier method is reported to be unstable with other semi-implicit methods (the analysis has not been carried out here). The nominal update is

$$u^{n+1} = u^n + (R^n_E + R^{n+1}_I)\Delta t. \tag{60}$$

Here R_I collects the stiff implicit terms — ion–neutral drag, collisional/frictional heating, and field-aligned heat conduction — and R_E collects the explicit terms (numerical fluxes plus the pressure-area and gravitational source).

Why split into explicit and implicit residuals. The partition is driven by numerical stiffness, not physical importance. The hyperbolic CFL constraint

$$\Delta t \lesssim C_{\text{CFL}} \frac{\Delta s}{|V| + c_s} \tag{61}$$

already controls the flux divergence. The pressure-area and gravity sources evolve on sound-crossing and free-fall timescales comparable to this limit, and the pressure-area term is in fact the geometric companion of the flux-divergence pressure-gradient force (Eq. (38) and the discussion that follows it). Keeping both in R_E is therefore consistent and stable at the acoustic timestep. The terms in R_I, by contrast, carry local relaxation rates

$$\lambda_{\text{drag}} = \alpha\Big(\tfrac{1}{\rho_i} + \tfrac{1}{\rho_n}\Big), \quad \lambda_{\text{temp}} = \frac{3k_B\alpha}{m_i + m_n}\Big(\tfrac{1}{C_i} + \tfrac{1}{C_n}\Big), \quad \lambda_{\text{cond}} \sim \frac{K}{C\,\Delta s^2}, \quad \lambda_{\text{ion}} = n_e[S_i(T_e) + \alpha_r(T_e)]\,, \tag{62}$$

that far exceed $1/\Delta t_{\text{CFL}}$ in regimes the model must cover: $\alpha\Delta t \sim 60$ at Model C7 conditions in the lower chromosphere, $\kappa_e \propto T_e^{5/2}$ makes the parabolic limit $\Delta t \lesssim C\,\Delta s^2/K$ severe near the transition region, and $n_e\alpha_r$ reaches $\gtrsim 1\ \text{s}^{-1}$ in the upper chromosphere/TR where the recombination time falls below the acoustic crossing time of one cell.

The two "pure" alternatives are both unattractive. A fully explicit update

$$u^{n+1} = u^n + \Delta t\,[R_E(u^n) + R_I(u^n)] \tag{63}$$

is mathematically simpler but is bounded by

$$\Delta t < \min\Big[\Delta t_{\text{CFL}},\, \lambda^{-1}_{\text{drag}},\, \lambda^{-1}_{\text{temp}},\, \lambda^{-1}_{\text{cond}}\Big]\,, \tag{64}$$

which in this problem is tens to hundreds of times smaller than the acoustic CFL limit. A fully implicit update $u^{n+1} = u^n + \Delta t\,[R_E(u^{n+1}) + R_I(u^{n+1})]$ removes that constraint, but couples all cells and all variables through a large nonlinear Jacobian (the Rusanov fluxes, wave speeds, and reconstructed primitives all depend on u^{n+1}), and adds numerical diffusion to the hyperbolic part for which the explicit Godunov-type discretization of R_E is already accurate and natural under the CFL condition. The IMEX split (Ascher et al., 1997; Pareschi and Russo, 2005) keeps the wave-transport part explicit and local, and treats only the genuinely stiff relaxation/diffusion blocks implicitly, so Δt is set by wave propagation rather than by collisions or diffusion.

Why not Picard. A naive fixed-point evaluation of $R_I(u^{n+1})$,

$$u^{n+1,k+1} = u^n + \Delta t\,(R_E^n + R_I(u^{n+1,k})), \tag{65}$$

is a contraction only when $\rho(\Delta t\,\partial R_I/\partial u) < 1$. But Δt here is set by the hyperbolic CFL, while R_I contains relaxation rates

$$\lambda_{\text{drag}} = \alpha\Big(\tfrac{1}{\rho_i} + \tfrac{1}{\rho_n}\Big), \quad \lambda_{\text{temp}} = \frac{3k_B\alpha}{m_i + m_n}\Big(\tfrac{1}{C_i} + \tfrac{1}{C_n}\Big), \quad \lambda_{\text{cond}} \sim \frac{K}{C\,\Delta s^2}, \quad \lambda_{\text{ion}} = n_e(S_i + \alpha_r), \tag{66}$$

any of which can satisfy $\Delta t\,\lambda \gg 1$ in the lower chromosphere ($\alpha\Delta t \sim 60$ at Model C7 conditions), the upper transition region (large $\kappa_e \propto T_e^{5/2}$ for λ_{cond}; large $n_e\alpha_r$ for λ_{ion}). Picard therefore amplifies errors by $\Delta t\lambda$ per pass and diverges. We replace it with direct backward-Euler solves for each stiff block.

Core idea of the implicit stages. Although R_I couples four physically distinct stiff processes, each process has a much simpler structure once the variables it does not touch are held fixed:

- *Ion–neutral drag* is a local relaxation of the velocity drift $w = V - U$ at each cell, with the center-of-mass velocity conserved.
- *Temperature equilibration* is a local relaxation of the temperature difference $\Delta T = T_i - T_n$ at each cell, with the heat-capacity-weighted mean temperature conserved.
- *Heat conduction* is a spatial diffusion problem for the temperature T_s that couples neighboring cells along the field line, and reduces to a tridiagonal linear system once discretized.
- *Ionization–recombination* is a local relaxation of the ionization fraction $f = \rho_i/(\rho_i + \rho_n)$ at each cell, with the total mass $\rho_i + \rho_n$, the center-of-mass momentum, and the total non-radiative energy (minus the $\Gamma_{\text{ion}}\chi_H$ radiative loss) conserved.

Each implicit stage therefore freezes the variables not involved in its own physics, isolates a small ODE (drag, equilibration, ionization) or diffusion equation (conduction), and applies backward Euler. The first three yield scalar closed-form updates (linear in w and ΔT; quadratic in f); the fourth yields a tridiagonal system. This isolate-and-relax structure is the reason the operator split below works.

Operator split. The implicit residual R_I, written out in Eq. (54), contains four physically disjoint blocks: the drag and frictional-heating terms in rows $(\rho_i V, \rho_n U, e_i, e_n)$, the temperature-difference terms $\pm 3k_B\alpha(T_n - T_i)/(m_i + m_n)$ in rows (e_i, e_n), the conduction divergences $\mathcal{C}_{s,i}$ in rows (e_i, e_n), and the ionization–recombination sources ($\pm m_i\Gamma_{\text{net}}$, $\pm m_i(\Gamma_{\text{ion}}U - \Gamma_{\text{rec}}V)$, $Q_{\text{ion}}^{e_i}$, $Q_{\text{ion}}^{e_n}$) in all six rows. We advance each block in turn within one Δt:

$$U^* = U^n + \Delta t\,R_E(U^n), \qquad \text{(explicit MUSCL+Rusanov step)} \tag{67}$$
$$U^{**} = \mathcal{D}_{\Delta t}(U^*), \qquad \text{(drag + frictional heating, point-implicit)} \tag{68}$$
$$U^{***} = \mathcal{T}_{\Delta t}(U^{**}), \qquad \text{(ion–neutral temperature equilibration, point-implicit)} \tag{69}$$
$$U^{****} = \mathcal{K}_{\Delta t}(U^{***}), \qquad \text{(heat conduction, tridiagonal solve)} \tag{70}$$
$$U^{n+1} = \mathcal{I}_{\Delta t}(U^{****}), \qquad \text{(ionization–recombination, point-implicit)} \tag{71}$$

Stage E is placed last so that the lagged rate coefficients $S_i(T_e)$ and $\alpha_r(T_e)$ are evaluated against the most up-to-date temperature produced by Stages B–D. The gravitational potential ϕ_g is unchanged by stages 68–71; the densities ρ_i and ρ_n are unchanged by stages 68–70 but *do* move in stage 71. Conserved energies are reconstructed at the end of each stage from $e = \frac{3}{2}p + \frac{1}{2}\rho v^2 + \rho\phi_g$, with the χ_H radiative loss subtracted in stage 71 as described below. The four implicit stages are derived in turn below by isolating the corresponding rows of $\mathbf{S}$, eliminating the now-static variables, and applying backward Euler to the resulting reduced system.

The two volumetric heat capacities used below are

$$C_i \equiv \left.\frac{\partial e_{\rm i,thermal}}{\partial T_i}\right|_{n_i} = 3\,n_i\,k_B, \qquad C_n \equiv \left.\frac{\partial e_{\rm n,thermal}}{\partial T_n}\right|_{n_n} = \tfrac{3}{2}\,n_n\,k_B, \tag{72}$$

where the factor of two in C_i reflects the electron contribution $p_i = 2n_ik_BT_i$ from quasi-neutrality.

How the backward-Euler formulas are obtained

Stages B and C both reduce, after a change of variables, to a scalar relaxation ODE of the form

$$\frac{dy}{dt} = -\lambda\,y, \tag{73}$$

where y is the deviation from a conserved mean — the velocity drift w in Stage B, the temperature difference ΔT in Stage C — and $\lambda > 0$ is a relaxation rate evaluated at the lagged state. Backward Euler applied to Eq. (73) replaces y on the right-hand side by its end-of-step value $y^{\rm new}$:

$$\frac{y^{\rm new} - y^{\rm old}}{\Delta t} = -\lambda\,y^{\rm new}, \tag{74}$$

which rearranges to a closed-form backward-Euler update

$$(1 + \Delta t\,\lambda)\,y^{\rm new} = y^{\rm old}, \qquad y^{\rm new} = \frac{y^{\rm old}}{1 + \Delta t\,\lambda}. \tag{75}$$

Equation (75) is not the exact continuous-time solution of Eq. (73) (that would be $y^{\rm old}e^{-\lambda\Delta t}$); rather, it is the closed-form update produced by the implicit Euler discretization. It is unconditionally stable for any $\Delta t\,\lambda \geq 0$ and recovers the fully-relaxed limit $y^{\rm new} \to 0$ as $\Delta t\,\lambda \to \infty$, which is precisely the regime that defeated the Picard iteration above. We use it with $(y, \lambda) = (w, \lambda_{\rm drag})$ in Stage B and $(y, \lambda) = (\Delta T, \lambda_{\rm temp})$ in Stage C. Stage D applies the same backward-Euler idea, but to a vector unknown coupled through a discrete diffusion operator, and therefore produces the tridiagonal system of Eq. (97) rather than a scalar formula.

Stage B: ion–neutral drag and frictional heating

Equations to solve. Starting from the Conservative Vector Form, we keep only the drag-related rows of the source vector $\mathbf{S}$, drop the flux divergence (Stage A has already advanced fluxes through U^*), and drop the explicit-source rows (pressure-area and gravity, in R_E) and the temperature-difference and conduction rows (handled in Stages C and D). Densities are frozen in this stage, so the local ODE system at each cell is

$$\tfrac{d}{dt}\rho_i = 0, \qquad \tfrac{d}{dt}\rho_n = 0, \tag{76}$$

$$\tfrac{d}{dt}(\rho_i V) = -\alpha(V - U), \qquad \tfrac{d}{dt}(\rho_n U) = +\alpha(V - U), \tag{77}$$

$$\tfrac{d}{dt}e_i = \tfrac{\alpha m_n}{m_i + m_n}(V - U)^2 - \alpha V(V - U), \qquad \tfrac{d}{dt}e_n = \tfrac{\alpha m_i}{m_i + m_n}(V - U)^2 + \alpha U(V - U). \tag{78}$$

With ρ_i and ρ_n held fixed by Eq. (76), the momentum rows reduce to a linear two-variable system for (V, U), and the energy rows decouple from this system up to a known forcing. Adding Eqs. (77) and Eqs. (78) gives two exact conservation laws,

$$\tfrac{d}{dt}(\rho_i V + \rho_n U) = 0, \qquad \tfrac{d}{dt}(e_i + e_n) = 0, \tag{79}$$

which we will enforce in the discrete update.

Reduction to a single relaxing variable. Rather than advancing (V, U) directly, we change variables to a part that is conserved by the drag and a part that relaxes. Introduce the drift, the center-of-mass velocity, and the reduced mass,

$$w = V - U, \qquad V_{\rm cm} = \frac{\rho_i V + \rho_n U}{\rho_i + \rho_n}, \qquad \mu = \frac{\rho_i \rho_n}{\rho_i + \rho_n}. \tag{80}$$

Adding the momentum rows of Eq. (77) gives $\dot{V}_{\rm cm} = 0$ exactly (Eq. (79)), while dividing them by ρ_i and ρ_n and subtracting yields a scalar relaxation equation for the drift:

$$\dot{w} = -\lambda_{\rm drag}\, w, \qquad \lambda_{\rm drag} \equiv \alpha\left(\tfrac{1}{\rho_i} + \tfrac{1}{\rho_n}\right), \qquad \dot{V}_{\rm cm} = 0. \tag{81}$$

$V_{\rm cm}$ and w have therefore decoupled the dynamics: $V_{\rm cm}$ is conserved exactly, and w obeys the scalar relaxation ODE Eq. (73) with $\lambda = \lambda_{\rm drag}$.

Backward-Euler solve. Freezing α at the lagged value $\alpha^* = \rho_i^* \nu_{in}(T_i^*, T_n^*, n_n^*)$ makes Eq. (81) a constant-coefficient linear ODE, and the scalar template Eq. (75) gives the closed-form backward-Euler update

$$w^{**} = \frac{w^*}{1 + \Delta t\, \lambda_{\rm drag}}, \qquad V_{\rm cm}^{**} = V_{\rm cm}^*. \tag{82}$$

The species velocities are then reconstructed from $(V_{\rm cm}^{**}, w^{**})$ by inverting Eq. (80):

$$V^{**} = V_{\rm cm}^* + \frac{\rho_n^*}{\rho_i^* + \rho_n^*}\, w^{**}, \qquad U^{**} = V_{\rm cm}^* - \frac{\rho_i^*}{\rho_i^* + \rho_n^*}\, w^{**}. \tag{83}$$

Frictional heating by energy bookkeeping. The drag step has contracted the drift from w^* to w^{**}, removing drift kinetic energy from the bulk flow. Because the original R_I rows conserve $e_i + e_n$ exactly (Eq. (79), right), that lost kinetic energy must reappear as ion and neutral thermal energy. Rather than time-integrating the nonlinear αw^2 source pieces in Eq. (78) directly — which can drift away from conservation in floating-point arithmetic — we compute the exact kinetic-energy deficit produced by Eq. (82) and deposit it into the thermal pools. Using $V_{\rm cm}^{**} = V_{\rm cm}^*$,

$$\Delta E_{\rm drag} = \tfrac{1}{2}\rho_i^*(V^*)^2 + \tfrac{1}{2}\rho_n^*(U^*)^2 - \tfrac{1}{2}\rho_i^*(V^{**})^2 - \tfrac{1}{2}\rho_n^*(U^{**})^2 = \tfrac{1}{2}\mu^*\left[(w^*)^2 - (w^{**})^2\right]. \tag{84}$$

The mass-weighted split of the frictional source rows in Eq. (78) sends the $\alpha m_n/(m_i + m_n)\, w^2$ piece to the ion thermal pool and the $\alpha m_i/(m_i + m_n)\, w^2$ piece to the neutral thermal pool. Matching this split against the total $\Delta E_{\rm drag}$ gives

$$\Delta E_{\rm i,fric} = \frac{m_n}{m_i + m_n}\, \Delta E_{\rm drag}, \qquad \Delta E_{\rm n,fric} = \frac{m_i}{m_i + m_n}\, \Delta E_{\rm drag}, \tag{85}$$

which feeds into the thermal pressures via $\Delta p_s = \frac{2}{3}\Delta E_{s,\rm fric}$ (from $e_{s,\rm thermal} = \frac{3}{2}p_s$):

$$p_i^{**} = p_i^* + \tfrac{2}{3}\Delta E_{\rm i,fric}, \qquad p_n^{**} = p_n^* + \tfrac{2}{3}\Delta E_{\rm n,fric}. \tag{86}$$

Conserved energies are then rebuilt from $(\rho, V^{**}, U^{**}, p^{**})$ using $e = \frac{3}{2}p + \frac{1}{2}\rho v^2 + \rho\phi_g$. By construction, $\rho_i V + \rho_n U$ and $e_i + e_n$ are exactly preserved at the discrete level.

Stage C: ion–neutral temperature equilibration

Equations to solve. After Stage B, the only R_I source rows still untreated outside of conduction are the $3k_B(T_n - T_i)$ collisional-exchange pieces of the energy equations in $\mathbf{S}$:

$$\tfrac{d}{dt}e_i = \frac{3k_B\alpha}{m_i + m_n}(T_n - T_i), \qquad \tfrac{d}{dt}e_n = \frac{3k_B\alpha}{m_i + m_n}(T_i - T_n). \tag{87}$$

Densities are frozen as before, and the kinetic-energy contribution to e_s is also frozen because V and U are unchanged in this stage. So only the thermal pieces of the energies evolve, and $de_s = de_{s,\rm thermal} = C_s\, dT_s$ with the volumetric heat capacities of Eq. (72). Equations (87) therefore reduce to a linear 2×2 ODE in temperatures alone,

$$C_i\, \dot{T}_i = \beta(T_n - T_i), \qquad C_n\, \dot{T}_n = \beta(T_i - T_n), \qquad \beta \equiv \frac{3k_B\alpha}{m_i + m_n}. \tag{88}$$

Adding the two rows yields $\frac{d}{dt}(C_i T_i + C_n T_n) = 0$, i.e. thermal energy is conserved by the collisional exchange.

Reduction to a single relaxing variable. The structure is exactly the same as in Stage B with $(V, U) \to (T_i, T_n)$ and $\rho_s \to C_s$. We again split (T_i, T_n) into a conserved mean and a relaxing deviation,

$$T_{\rm cm} = \frac{C_i T_i + C_n T_n}{C_i + C_n}, \qquad \Delta T = T_i - T_n, \tag{89}$$

so that $T_{\rm cm}$ plays the role $V_{\rm cm}$ played in Stage B, and ΔT plays the role of w. Adding and subtracting the rows of Eq. (88) after dividing by C_i and C_n gives $\dot{T}_{\rm cm} = 0$ and a scalar relaxation equation for ΔT:

$$\dot{\Delta T} = -\lambda_{\rm temp}\, \Delta T, \qquad \lambda_{\rm temp} \equiv \beta\Big(\tfrac{1}{C_i} + \tfrac{1}{C_n}\Big), \qquad \dot{T}_{\rm cm} = 0. \tag{90}$$

Backward-Euler solve. With β lagged at the post-drag state, Eq. (90) is a constant-coefficient scalar relaxation, and the template Eq. (75) again gives a closed-form update,

$$\Delta T^{***} = \frac{T_i^{**} - T_n^{**}}{1 + \Delta t\, \lambda_{\rm temp}}, \qquad T_{\rm cm}^{***} = T_{\rm cm}^{**} = \frac{C_i T_i^{**} + C_n T_n^{**}}{C_i + C_n}. \tag{91}$$

The species temperatures are reconstructed by inverting Eq. (89):

$$T_i^{***} = T_{\rm cm}^{**} + \frac{C_n}{C_i + C_n}\, \Delta T^{***}, \qquad T_n^{***} = T_{\rm cm}^{**} - \frac{C_i}{C_i + C_n}\, \Delta T^{***}. \tag{92}$$

Pressures are rebuilt by $p_i^{***} = 2 n_i k_B T_i^{***}$ and $p_n^{***} = n_n k_B T_n^{***}$, and conserved energies from $(\rho, V^{**}, U^{**}, p^{***})$.

Stage D: tridiagonal heat conduction

Why this stage is different. Stages B and C are point-implicit: the ODE at each cell involves only that cell's own variables, so the backward-Euler update is local and reduces to the scalar template Eq. (75). Heat conduction is fundamentally different. The conduction divergence in the energy equation involves T_s at the neighboring cells $i \pm 1$ as well as at cell i, so the backward-Euler discretization couples the unknowns $T_{s,i-1}^{n+1}, T_{s,i}^{n+1}, T_{s,i+1}^{n+1}$. The result is no longer a scalar formula but a linear system; because the spatial stencil has three points, that system is tridiagonal and still solvable at $O(N)$ cost by the Thomas algorithm.

Equations to solve. The remaining R_I source rows are the conduction divergences of the energy equations in $\mathbf{S}$:

$$\tfrac{\partial}{\partial t} e_i = B\, \tfrac{\partial}{\partial s}\left[\frac{K_i}{B}\, \tfrac{\partial T_i}{\partial s}\right], \qquad \tfrac{\partial}{\partial t} e_n = B\, \tfrac{\partial}{\partial s}\left[\frac{\kappa_n}{B}\, \tfrac{\partial T_n}{\partial s}\right], \tag{93}$$

with $K_i = \kappa_i + \kappa_e(T_i, n_i, n_n)$. Densities and velocities are frozen, so as in Stage C $de_s = C_s\, dT_s$, and Eq. (93) becomes a linear parabolic equation in T_s alone:

$$C_s\, \tfrac{\partial T_s}{\partial t} = B\, \tfrac{\partial}{\partial s}\left[\frac{K_s}{B}\, \tfrac{\partial T_s}{\partial s}\right] \quad \text{for } s \in \{i, n\}. \tag{94}$$

Spatial discretization. The right-hand side of Eq. (94) is the same operator already discretized in Eqs. (55)–(56),

$$\mathcal{C}_{s,i}(\mathbf{T}_s) = \frac{B_i}{\Delta s_i}\left[\frac{K_{s,i+1/2}}{B_{i+1/2}}\, \frac{T_{s,i+1} - T_{s,i}}{\Delta s_{i+1/2}} - \frac{K_{s,i-1/2}}{B_{i-1/2}}\, \frac{T_{s,i} - T_{s,i-1}}{\Delta s_{i-1/2}}\right]. \tag{95}$$

The right-hand side of Eq. (95) depends linearly on three temperatures: $T_{s,i-1}$, $T_{s,i}$, and $T_{s,i+1}$.

Backward-Euler discretization. Apply backward Euler in time to Eq. (94) with the spatial operator from Eq. (95) and conductivities lagged at T_s^{***}:

$$C_{s,i}\, \frac{T_{s,i}^{n+1} - T_{s,i}^{***}}{\Delta t} = \mathcal{C}_{s,i}(\mathbf{T}_s^{n+1}). \tag{96}$$

Substituting Eq. (95) into the right-hand side of Eq. (96) and collecting the coefficients of $T_{s,i-1}^{n+1}$, $T_{s,i}^{n+1}$, and $T_{s,i+1}^{n+1}$ produces, at each interior cell i, the tridiagonal row

$$\boxed{-L_i\,T_{s,i-1}^{n+1} + \big(1 + L_i + R_i\big)T_{s,i}^{n+1} - R_i\,T_{s,i+1}^{n+1} = T_{s,i}^{***},} \tag{97}$$

where the dimensionless lagged coefficients are

$$L_i = \frac{\Delta t}{C_{s,i}}\,\frac{B_i}{\Delta s_i}\,\frac{K_{s,i-1/2}}{B_{i-1/2}\,\Delta s_{i-1/2}}, \qquad R_i = \frac{\Delta t}{C_{s,i}}\,\frac{B_i}{\Delta s_i}\,\frac{K_{s,i+1/2}}{B_{i+1/2}\,\Delta s_{i+1/2}}. \tag{98}$$

By construction L_i is exactly the prefactor of $T_{s,i-1}^{n+1}$ and R_i is exactly the prefactor of $T_{s,i+1}^{n+1}$ that arise from this collection; the diagonal entry $1 + L_i + R_i$ is then forced by the ∂_t term on the left of Eq. (96) and by the fact that the conduction stencil sums to zero on constant temperature. The charged-fluid row uses $K_s = \kappa_i + \kappa_e(T_i, n_i, n_n)$, the neutral row uses $K_s = \kappa_n$.

Boundary conditions. The ghost temperatures $T_{s,-1}, T_{s,N}$ are Dirichlet data (computed from the boundary cons vectors with the same ϕ_g extrapolation used in Eqs. (55)–(56)). They are moved to the right-hand side of Eq. (97): at $i = 0$,

$$(1 + L_0 + R_0)T_{s,0}^{n+1} - R_0\,T_{s,1}^{n+1} = T_{s,0}^{***} + L_0\,T_{s,-1}, \tag{99}$$

and at $i = N - 1$,

$$-L_{N-1}\,T_{s,N-2}^{n+1} + (1 + L_{N-1} + R_{N-1})T_{s,N-1}^{n+1} = T_{s,N-1}^{***} + R_{N-1}\,T_{s,N}. \tag{100}$$

The two tridiagonal systems (one per species) are solved by the Thomas algorithm; pressures are then rebuilt from the new temperatures via $p_i^{n+1} = 2n_i k_B T_i^{n+1}$ and $p_n^{n+1} = n_n k_B T_n^{n+1}$, and conserved energies from $(\rho, V^{**}, U^{**}, p^{****})$.

Stage E: ionization–recombination

Equations to solve. After Stage D, the only R_I rows still untreated are the ionization–recombination sources from §2.3: $\pm m_i\Gamma_{\rm net}$ on the mass rows, $\pm m_i(\Gamma_{\rm ion}U - \Gamma_{\rm rec}V)$ on the momentum rows, and $Q_{\rm ion}^{e_i}$, $Q_{\rm ion}^{e_n}$ on the energy rows. The kinetic-energy pieces of the energy sources are local bookkeeping (they follow from the momentum sources); the rest form a scalar ODE for the ionization fraction $f \equiv \rho_i/\rho_{\rm tot}$ with $\rho_{\rm tot} = \rho_i + \rho_n$ held fixed.

Reduction to a scalar ODE. Using $n_e = n_i = \rho_{\rm tot} f/m_i$, $n_n = \rho_{\rm tot}(1 - f)/m_n$ with $m_i = m_n = m$ for hydrogen, and defining $n_{\rm tot} \equiv \rho_{\rm tot}/m$, the mass-source rows of R_I reduce, after dividing by $\rho_{\rm tot}$, to

$$\dot f = n_{\rm tot}\big[\, f(1 - f)\,S_i(T_e) - f^2\,\alpha_r(T_e)\,\big] + (1 - f)\,P_{\rm phot} \equiv G(f, T_e), \tag{101}$$

where the photoionization term $(1-f)P_{\rm phot}$ comes from $\Gamma_{\rm phot} = n_n P_{\rm phot}$ divided by $\rho_{\rm tot}$ (see Eq. (130)). Conservation of $\rho_{\rm tot}$, of the center-of-mass momentum $\rho_i V + \rho_n U$, and (modulo the χ_H drain) of total energy follows by inspection of the sources. As in Stages B and C, the ODE decouples a conserved mean ($\rho_{\rm tot}$, $V_{\rm cm}$, $T_{\rm cm}$, etc.) from a single relaxing variable, here f.

Backward-Euler quadratic solve. Lagging the rate coefficients at the post-conduction temperature $T_e^{****} = T_i^{****}$ makes Eq. (101) a constant-coefficient ODE quadratic in f. Backward Euler gives

$$f^{n+1} - f^{****} = \Delta t\big\{n_{\rm tot}^{****}\big[f^{n+1}(1 - f^{n+1})S_i - (f^{n+1})^2\alpha_r\big] + (1 - f^{n+1})P_{\rm phot}\big\}, \tag{102}$$

which rearranges to the quadratic

$$A\,(f^{n+1})^2 + B\,f^{n+1} - C = 0, \tag{103}$$

$$\begin{aligned} A &\equiv \Delta t\,n_{\rm tot}^{****}\big(S_i + \alpha_r\big),\\ B &\equiv 1 - \Delta t\,n_{\rm tot}^{****}\,S_i + \Delta t\,P_{\rm phot},\\ C &\equiv f^{****} + \Delta t\,P_{\rm phot}. \end{aligned} \tag{104}$$

The physical root is the unique positive root in $[0, 1]$,

$$f^{n+1} = \frac{-B + \sqrt{B^2 + 4A\,C}}{2A} \quad (A > 0), \tag{105}$$

with the limit $f^{n+1} \to C/B$ recovered as $A \to 0$ (collisional rates vanish; the pure photoionization/recombination limit). The discriminant is strictly non-negative for $A, C, P_{\text{phot}} \geq 0$, so the solve is unconditionally well-posed. A scalar Newton fallback on Eq. (102) is retained for robustness in the limit of very small A.

Integrated rates and mass update. Once f^{n+1} is known, the *integrated* ionizations and recombinations during this Δt are

$$\begin{aligned} \Gamma^{\Delta}_{\text{coll}} &= \Delta t\, n_e^{n+1}\, n_n^{n+1}\, S_i(T_e^{****}), \quad \Gamma^{\Delta}_{\text{phot}} = \Delta t\, n_n^{n+1}\, P_{\text{phot}}, \\ \Gamma^{\Delta}_{\text{ion}} &= \Gamma^{\Delta}_{\text{coll}} + \Gamma^{\Delta}_{\text{phot}}, \qquad \Gamma^{\Delta}_{\text{rec}} = \Delta t\, (n_e^{n+1})^2\, \alpha_r(T_e^{****}), \end{aligned} \tag{106}$$

evaluated at the post-step densities $n_e^{n+1} = n_i^{n+1} = n_{\text{tot}} f^{n+1}$ and $n_n^{n+1} = n_{\text{tot}}(1 - f^{n+1})$, with rate coefficients lagged at the post-conduction temperature. By construction $\Gamma^{\Delta}_{\text{ion}} - \Gamma^{\Delta}_{\text{rec}} = (f^{n+1} - f^{****})\, n_{\text{tot}}$, equal to $\Delta n_i/m$; a tiny round-off mismatch from the lagged-vs-implicit evaluation of T_e is redistributed in proportion to $\Gamma^{\Delta}_{\text{ion}}/(\Gamma^{\Delta}_{\text{ion}} + \Gamma^{\Delta}_{\text{rec}})$ to enforce the discrete identity exactly. Densities update as

$$\rho_i^{n+1} = \rho_{\text{tot}} f^{n+1}, \qquad \rho_n^{n+1} = \rho_{\text{tot}}(1 - f^{n+1}). \tag{107}$$

Momentum and energy bookkeeping. The species momenta and total energies are updated using $\Gamma^{\Delta}_{\text{ion}}$ and $\Gamma^{\Delta}_{\text{rec}}$ at the lagged velocities V^{**}, U^{**} (unchanged by Stages C and D) and the lagged temperatures T_i^{****}, T_n^{****}:

$$(\rho_i V)^{n+1} = (\rho_i V)^{****} + m\big(\Gamma^{\Delta}_{\text{ion}} U^{**} - \Gamma^{\Delta}_{\text{rec}} V^{**}\big)\,, \tag{108}$$

$$(\rho_n U)^{n+1} = (\rho_n U)^{****} - m\big(\Gamma^{\Delta}_{\text{ion}} U^{**} - \Gamma^{\Delta}_{\text{rec}} V^{**}\big)\,, \tag{109}$$

$$\begin{aligned} e_i^{n+1} = e_i^{****} &+ \Gamma^{\Delta}_{\text{ion}}\big(\tfrac{3}{2} k_B T_n^{****} + \tfrac{1}{2} m (U^{**})^2\big) \\ &- \Gamma^{\Delta}_{\text{rec}}\big(\tfrac{3}{2} k_B T_i^{****} + \tfrac{1}{2} m (V^{**})^2\big) - \Gamma^{\Delta}_{\text{coll}}\, \chi_H, \end{aligned} \tag{110}$$

$$\begin{aligned} e_n^{n+1} = e_n^{****} &- \Gamma^{\Delta}_{\text{ion}}\big(\tfrac{3}{2} k_B T_n^{****} + \tfrac{1}{2} m (U^{**})^2\big) \\ &+ \Gamma^{\Delta}_{\text{rec}}\big(\tfrac{3}{2} k_B T_i^{****} + \tfrac{1}{2} m (V^{**})^2\big)\,. \end{aligned} \tag{111}$$

Total mass $\rho_i + \rho_n$ and total momentum $(\rho_i V + \rho_n U)$ are conserved exactly by Eqs. (107)–(109). The combined energy update obeys $e_i^{n+1} + e_n^{n+1} - (e_i + e_n)^{****} = -\Gamma^{\Delta}_{\text{coll}}\, \chi_H$, the optically-thin Lyman-continuum loss anticipated by Eq. (15). Photoionizations contribute to $\Gamma^{\Delta}_{\text{ion}}$ (mass + thermal/KE inheritance bookkeeping) but *not* to the χ_H drain (Carlsson and Stein, 2002).

χ_H floor and the radiative-heating regime. In a regime where $\Gamma^{\Delta}_{\text{coll}} \chi_H$ exceeds the available ion thermal energy $\frac{3}{2} \cdot 2 n_i k_B T_i\, \Delta s$, the naive update Eq. (110) would drive $T_e < 0$. We floor the post-step electron pressure at a small positive value $p_{i,\min} = 2 n_i k_B T_{\text{floor}}$ and log a warning. This floor is *not* conservative: the energy that is "floored away" is real energy lost to an unphysical regime, signaling that radiative heating (absent in this model) must in reality replace the χ_H drain in the upper chromosphere. The photoionization term added in this revision recovers a non-trivial chromospheric ionization fraction but does *not* act as a heating source; an explicit radiative heating term following Carlsson and Leenaarts (2012) remains a follow-on.

Algorithmic cost and stability. Stages B, C, and E are local: $O(N)$ work, no allocations beyond per-cell scratch. Stage D is two Thomas sweeps, also $O(N)$. The composite map is unconditionally stable for each stiff piece in isolation; the only remaining Δt constraint is the hyperbolic CFL of Eq. (67). For nonlinear $\kappa_e \propto T_e^{5/2}$, the lagged-conductivity choice introduces an $O(\Delta t)$ linearization error; if greater accuracy is required, a Picard wrapper around stage D — recomputing $K(T)$ and re-solving Eq. (97) — converges quickly because each inner sweep is itself an implicit solve, not a fixed point on the stiff operator. The same comment applies to the lagged-rate choice in stage E: a Picard wrapper recomputing S_i and α_r at the new T_e and re-solving Eq. (104) would tighten the χ_H-drain/temperature feedback in the transition region.

5 Closures: Heat Conductivities and Ionization Rates

5.1 Electron Heat Conductivity

In this subsection, we derive the electron heat conductivity in a partially ionized plasma by incorporating the electron–neutral collision frequency.

We begin with Spitzer's collisional formulation (Spitzer, 1962):

$$\kappa_e^{\mathrm{Sp}} = (4\pi\epsilon_0)^2 \frac{20 m_e^2 k_B v_{th}^5}{3e^4 Z \ln\Lambda} \left(\frac{2}{3\pi}\right)^{\frac{3}{2}} \delta_T \epsilon, \tag{112}$$

where ϵ_0, m_e, k_B, v_{th}, and e are vacuum permittivity, electron mass, Boltzmann constant, root-mean-square velocity, and electron charge. The parameters ϵ and δ_T are dimensionless coefficients; for $Z = 1$ they take the values

$$\delta_T = 0.2252, \quad \epsilon = 0.4189. \tag{113}$$

We next account for electron–neutral collisions by writing the thermal conductivity in a more general fluid form,

$$\kappa \propto k_B n_e v_{th} \lambda_{\mathrm{mfp}} \sim \frac{k_B n_e v_{th}^2}{\nu} \sim \frac{k_B^2 n_e T_e}{m_e \nu}, \tag{114}$$

where λ_{mfp} and ν are the mean free path and collision frequency, respectively.

In Spitzer's formulation, electron–ion and electron–electron collisions are considered with frequencies approximated as

$$\nu_{ee} \approx \nu_{ei} = \frac{n_e e^4}{4\pi\epsilon_0^2 \sqrt{k_B^2 m_e T_e^3}} \ln\Lambda. \tag{115}$$

Since $\nu = \nu_{ei} + \nu_{ee} \approx 2\nu_{ei}$, the electron heat conductivity becomes

$$\kappa_e^{\mathrm{Sp}} = 320 \sqrt{\frac{2}{\pi}} \delta_T \epsilon \frac{k_B^2 n_e T_e}{m_e \nu}. \tag{116}$$

To include electron–neutral collisions, we replace ν with the total collision frequency $\nu_{\mathrm{tot}} = \nu_{ei} + \nu_{ee} + \nu_{en}$, where the electron–neutral collision frequency is

$$\nu_{en} = n_n \sigma_{\mathrm{en}} \int v f_e(v)\,\mathrm{d}v = n_n \sigma_{\mathrm{en}} \sqrt{\frac{8 k_B T_e}{\pi m_e}} = n_n r_n^2 \sqrt{\frac{8\pi k_B T_e}{m_e}} = 5.5 \times 10^{-17} n_n \sqrt{T_e} \quad \text{(in SI units)}, \tag{117}$$

with $\sigma_{\mathrm{en}} \approx \pi r_n^2$ the electron–neutral cross section, r_n the neutral particle radius, and $f_e(v)$ the Maxwellian electron velocity distribution. This value, however, may not be accurate; we instead use $\nu_{en} = 1.95 \times 10^{-16} n_n \sqrt{T_e}$.

The modified electron heat conductivity is then

$$\kappa_e(T_e, n_e, n_n) = 320 \sqrt{\frac{2}{\pi}} \delta_T \epsilon \frac{k_B^2 n_e T_e}{m_e \nu_{\mathrm{tot}}}, \tag{118}$$

or, in SI units,

$$\kappa_e(T_e, n_e, n_n) = \frac{9.2048 \times 10^{-12}\, n_e\, T_e^{5/2}}{n_e + 3.5609 \times 10^{-12}\, n_n\, T_e^2}. \tag{119}$$

For $T_e = T_n = 6500$ K,

$$\kappa_e(6500, n_e, n_n) = \frac{0.0313545 n_e}{n_e + 0.00015045 n_n}. \tag{120}$$

Only when $n_n \sim 1000\, n_e$ do electron–neutral collisions contribute significantly to κ_e. For $n_n \sim n_e$, $\kappa_e \approx 0.031$ W/(m·K).

5.2 Neutral Heat Conduction

The thermal conductivity of a neutral gas is

$$\kappa_n = \frac{1}{3} c_v \rho \bar{c} \lambda_{\mathrm{mfp}}, \tag{121}$$

where c_v, ρ, $\bar{c}$, and λ_{mfp} are the heat capacity per unit mass, mass density, mean velocity, and mean free path.

p, бар → / T, °K	10^{-5}	10^{-4}	10^{-3}	10^{-2}	0,1	1	10	100	200
1400	0,7222	0,6304	0,6013	0,5921	0,5892	0,5883	0,5880	0,5879	0,5879
1600	1,951	1,071	0,7910	0,7021	0,6739	0,6650	0,6622	0,6613	0,6612
1800	7,848	3,080	1,485	0,9724	0,8095	0,7578	0,7415	0,7363	0,7356
2000	23,07	9,687	3,795	1,770	1,114	0,9050	0,8387	0,8178	0,8149
2200	24,72	22,12	9,589	3,834	1,838	1,190	0,9829	0,9174	0,9085
2400	7,330	23,93	19,07	8,149	3,381	1,745	1,215	1,047	1,024
2600	2,154	9,991	23,51	14,87	6,203	2,755	1,597	1,224	1,173
2800	1,285	3,347	14,80	20,75	10,49	4,418	2,203	1,474	1,374
3000	1,165	1,711	6,255	19,43	15,49	6,871	3,115	1,827	1,649
3200	1,186	1,349	2,890	12,12	18,78	10,02	4,396	2,311	2,018
3400	1,235	1,291	1,837	6,269	17,72	13,40	6,067	2,951	2,500
3600	1,291	1,312	1,523	3,456	13,09	16,00	8,077	3,763	3,110
3800	1,346	1,355	1,444	2,301	8,326	16,68	10,26	4,747	3,853
4000	1,402	1,406	1,447	1,847	5,185	15,05	12,31	5,887	4,725
4200	1,457	1,459	1,479	1,677	3,477	11,99	13,87	7,145	5,712
4400	1,512	1,513	1,523	1,627	2,612	8,849	14,56	8,461	6,785
4600	1,566	1,567	1,572	1,630	2,186	6,387	14,22	9,748	7,902
4800	1,620	1,621	1,624	1,657	1,983	4,715	12,97	10,89	8,996
5000	1,675	1,675	1,677	1,697	1,895	3,663	11,19	11,80	10,01
5200	1,730	1,730	1,731	1,744	1,868	3,022	9,297	12,38	10,87
5400	1,785	1,785	1,786	1,794	1,875	2,642	7,596	12,57	11,52
5600	1,841	1,841	1,842	1,847	1,901	2,421	6,214	12,37	11,90
5800	1,898	1,898	1,899	1,903	1,939	2,300	5,158	11,84	12,01
6000	1,957	1,957	1,958	1,960	1,986	2,241	4,384	11,05	11,83

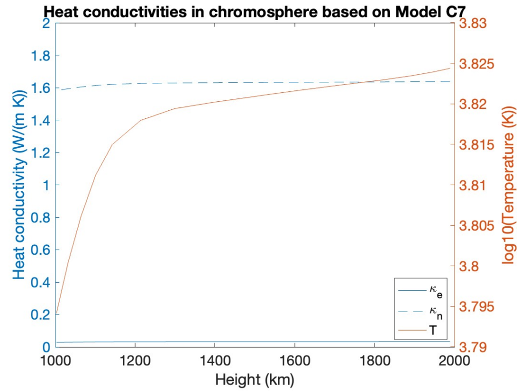

Figure 1: A comparison between experimental hydrogen thermal conductivity and the one obtained from (125). From the table it can be extrapolated that in the chromosphere where $T \approx 6000$ K and $n \approx 2 \times 10^{16}$ m^{-3}, the hydrogen gas, which is mostly composed of atomic hydrogen, has the thermal conductivity around 2 W/(m K). On the other hand, if we neglect ions, the calculation from (125) gives approximately 1.6 W/(m K).

For H gas, $\rho c_v = \left(\frac{\partial u}{\partial T}\right)_V = \frac{3}{2} n_n k_B$, so

$$\kappa_n = \frac{1}{2} n_n k_B \frac{\bar{c}^2}{\nu} = \frac{4}{\pi} \frac{k_B^2 n_n T_n}{m_n \nu}, \tag{122}$$

where $\nu = \nu_{ni} + \nu_{nn}$ and $\bar{c}$ is replaced by $\sqrt{8 k_B T_n / (\pi m_n)}$ from the Maxwellian distribution.

Approximating $m_i \approx m_n$, the collision frequencies ν_{ni} and ν_{nn} follow from the Maxwellian distribution of relative velocities, with relative velocities characterized by $v_{ni}^{\mathrm{rel}} \sim f(T_i + T_n)$ and $v_{nn}^{\mathrm{rel}} \sim f(2T_n)$:

$$\nu_{ni} = n_i \sigma_{\mathrm{ni}} \sqrt{\frac{8 k_B (T_i + T_n)}{\pi m_n}} = n_i (r_i + r_n)^2 \sqrt{\frac{8 \pi k_B (T_i + T_n)}{m_n}}, \tag{123}$$

$$\nu_{nn} = 4 n_n \sigma_{\mathrm{nn}} \sqrt{\frac{k_B T_n}{\pi m_n}} = 4 n_n (2 r_n)^2 \sqrt{\frac{\pi k_B T_n}{m_n}}. \tag{124}$$

Then, in SI units,

$$\kappa_n = \frac{0.0342006 n_n T_n}{1.20613 n_i \sqrt{T_n + T_i} + 1.70573 n_n \sqrt{T_n}}. \tag{125}$$

For $T_n = T_i = 6500$ K and $n_i = n_n$,

$$\kappa_n = 0.947 \quad \mathrm{W/(m \cdot K)}. \tag{126}$$

This is far greater than κ_e at the same temperature. Only above $\sim 10^5$ K does $\kappa_n = 3.71$ W/(m·K) become an order of magnitude smaller than $\kappa_e = 29.1$ W/(m·K).

5.3 Validation and Application to Model C7

A comparison between experimental hydrogen thermal conductivity and (125) is shown in Figure 1.

Applying the heat-conductivity formulation to Model C7 — whose plasma profile (ion density, total hydrogen density, atomic hydrogen density, and temperature) is shown in Figure 2 — we find that neutral heat conductivity dominates over electron heat conductivity throughout the chromosphere. The roles switch in the transition region, where the temperature ramps up sharply.

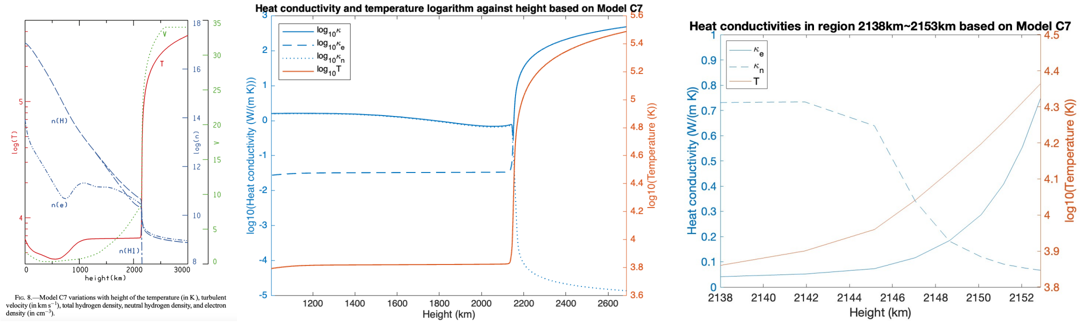

Figure 2: **Left**: Model C7 profile, including ion density, total hydrogen density, atomic hydrogen density, and temperature. **Middle**: heat conductivity logarithms and temperature logarithm against height calculated from the profile. **Right**: heat conductivities and temperature logarithms against height (zoomed in). Neutral heat conductivity dominates over electron in the chromosphere; their roles switch in the transition region. The two conductivities intersect at height = 2149 km.

5.4 Fitted Piecewise Model

We evaluate the total conductivity $\kappa = \kappa_n + \kappa_e$. Because Model C7 has a monotonic temperature–height relation, κ can be expressed as a function of temperature alone (Figure 3). A notable drop appears in the temperature range $[6400\text{ K}, 6700\text{ K}]$, driven by sharp changes in ionization where the temperature varies only weakly.

The total conductivity $\kappa(T)$ splits into a low-temperature region ($T < 2.5 \times 10^4$ K) dominated by κ_n and a high-temperature region ($T > 2.5 \times 10^4$ K) dominated by κ_e. We fit the low-temperature branch by a constant 0.89 W/(m·K), the average conductivity there, and the high-temperature branch by Spitzer's κ_e^{Sp}. The cross-over temperature is chosen so that the piecewise fit is continuous:

$$\kappa^{\text{fitted}}(T) = \begin{cases} 0.89\,\text{W}/(\text{m}\cdot\text{K}), & \text{if } T < 2.5 \times 10^4\,\text{K} \\ \kappa_{e,0}T^{5/2}\,\text{W}/(\text{m}\cdot\text{K}), & \text{if } T \geq 2.5 \times 10^4\,\text{K} \end{cases}$$

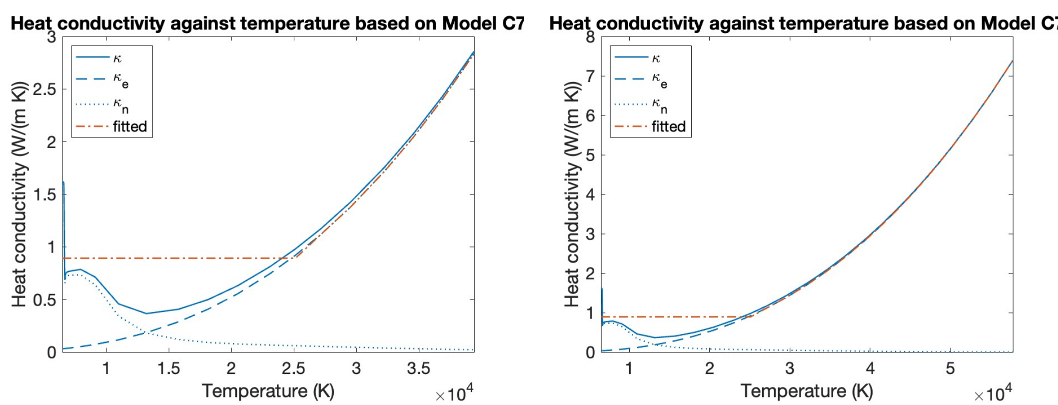

Figure 3: The heat conductivities κ_e, κ_n, and their sum $\kappa = \kappa_e + \kappa_n$ as functions of temperature, based on Model C7 (**left**: zoomed in; **right**: zoomed out). The total conductivity $\kappa(T)$ splits into a low-temperature region ($T < 2.5 \times 10^4$ K), where κ_n dominates, and a high-temperature region ($T > 2.5 \times 10^4$ K), where κ_e dominates. The fit uses a constant 0.89 W/(m·K) for the low-temperature region and κ_e^{Sp} for the high-temperature region, with the cross-over chosen to make the piecewise fit continuous.

5.5 Ionization and Recombination Rate Coefficients

The ionization-recombination block of §2.3 requires closure forms for $S_i(T_e)$ and $\alpha_r(T_e)$. We adopt standard atomic-data fits.

Electron-impact ionization (Voronov, 1997). Voronov (1997) gives a practical fit for the electron-impact ionization rate coefficient of atoms and ions $Z = 1$–28. For neutral hydrogen (H I) it reads, in CGS,

$$S_i(T_e) = A\,\frac{1 + P\sqrt{U}}{X + U}\,U^K\,\exp(-U) \quad [\mathrm{cm}^3/\mathrm{s}], \qquad U \equiv \chi_H/(k_B T_e), \tag{127}$$

with $A = 0.291 \times 10^{-7}$ cm^3/s, $P = 0$, $X = 0.232$, $K = 0.39$. Converting to SI (1 cm^3 = 10^{-6} m^3) gives the form used in the code,

$$S_i(T_e) = 2.91 \times 10^{-14}\,\frac{U^{0.39}\,\exp(-U)}{0.232 + U} \quad [\mathrm{m}^3/\mathrm{s}], \qquad U = \chi_H/(k_B T_e). \tag{128}$$

Eq. (128) reproduces the experimental cross-section data integrated over a Maxwellian electron distribution to better than $\sim 10\%$ for 1–50 eV.

Radiative recombination (case-B, Hummer, 1994). For the recombination rate we use a power-law fit to the Hummer (1994) case-B coefficient (see also Verner and Ferland 1996 for a complementary tabulation),

$$\alpha_r(T_e) = 2.7 \times 10^{-19}\left(\frac{T_e}{10^4\ \mathrm{K}}\right)^{-0.75} \quad [\mathrm{m}^3/\mathrm{s}], \tag{129}$$

accurate to $\sim 10\%$ for $T_e \in [10^3, 10^5]$ K, which covers the full range needed by Model C7. We choose case-B (no return to the ground state) rather than case-A because the optically-thin χ_H drain adopted in §2.3 is internally consistent only with case-B: case-A would double-count ground-state recombinations as immediately re-ionized by the same Lyman-continuum photon, which is correct only for a fully optically-thick layer. Snow and Hillier (2020) discusses this consistency choice in detail.

Effective photoionization rate (Carlsson and Stein, 2002; Leenaarts et al., 2007). The collisional Voronov rate S_i is exponentially suppressed at chromospheric temperatures ($n_e S_i \sim 10^{-5}$ s^{-1} at $T_e \sim 6500$ K; see "Timescales" below) and would by itself produce a fully recombined chromosphere ($f \to 0$) within hundreds of recombination times. The solar chromosphere is observed to maintain $f \sim 10^{-3}$–10^{-1} (Avrett and Loeser, 2008) because of radiative ionization. Carlsson and Stein (2002) show, using detailed non-LTE radiation-hydrodynamic simulations, that the dominant chromospheric ionization pathway is the two-step process *collisional excitation in Lyα ($n = 1 \to 2$) followed by photoionization in the Balmer continuum ($n = 2 \to$ continuum)*, while direct ground-state Lyman-continuum photoionization is negligible. The same paper reports that the corresponding ionization/recombination timescale ranges over 10^3–10^5 s throughout the chromosphere, shortening to seconds inside shocks. Leenaarts et al. (2007) encode this pathway via the "fixed radiative rates" approximation: precompute the radiative coupling on a reference atmosphere and inject it as a constant (or height-dependent) rate into the time-dependent rate equations.

For the two-state (ground + continuum) reduction adopted in §2.3 we summarize the two-step pathway by a single effective ground-state photoionization rate P_{phot} [s^{-1}] applied to the neutral population,

$$\Gamma_{\mathrm{phot}} = n_n\,P_{\mathrm{phot}}, \qquad P_{\mathrm{phot}}^{\mathrm{default}} = 1.0 \times 10^{-4}\ \mathrm{s}^{-1}. \tag{130}$$

The default value is the geometric mean of the Carlsson and Stein (2002) chromospheric timescale range; in the code it is exposed as the configurable scalar `grid.photoionization_rate`. A height-dependent profile $P_{\mathrm{phot}}(s)$, e.g. following the cooling/heating recipes of Carlsson and Leenaarts (2012), replaces the constant body of the `photoionization_rate_P` helper without changing any of the equations of §2.3.

We emphasize that this rate is a phenomenological closure: a faithful calculation would solve the radiative transfer for Lyα and the Balmer continuum self-consistently (Leenaarts et al., 2007). The single-rate closure used here is the minimum extension that converts an otherwise unphysically recombining chromosphere into one with a non-trivial ionization fraction; quantitative agreement with C7 in §6 requires the height-dependent generalization.

Timescales in the Model C7 atmosphere. At $T_e \sim 6500$ K, $U \sim 24$, so $S_i \sim 5\times10^{-23}$ m^3/s, and the local *collisional* ionization frequency $n_e S_i \sim 10^{-5}$ s^{-1} gives a collisional ionization time of days — far longer than any dynamical time in the chromosphere, and the reason a Voronov-only closure drives $f \to 0$ in the cool, low-density mid-chromosphere. The radiative ionization channel discussed above operates with $P_{\rm phot} \sim 10^{-4}$ s^{-1}, so radiative ionizations dominate collisional ones by ~ 10 at these temperatures. The recombination coefficient is $\alpha_r \sim 4 \times 10^{-19}$ m^3/s, so $n_e\alpha_r \sim 7 \times 10^{-2}$ s^{-1}, a recombination time of ~ 15 s. In the transition region $T_e \sim 10^4$ K, $U \sim 16$, S_i rises by orders of magnitude, and the recombination rate $n_e\alpha_r$ reaches ~ 1 s^{-1}. Both rates therefore satisfy $\lambda_{\rm ion}\Delta t_{\rm CFL} \gtrsim 1$ over part of the domain, motivating the implicit Stage E of §4.4.

Kinetic equilibrium vs. Model C7. With photoionization added (Eq. (130)), the chromospheric kinetic equilibrium of Eq. (16) gives $f_{\rm eq} \sim \sqrt{P_{\rm phot}/(n_{\rm tot}\alpha_r)}$, of order 10^{-3} for the mid-chromosphere C7 values ($n_{\rm tot} \sim 10^{19}$ m^{-3}, $\alpha_r \sim 4 \times 10^{-19}$ m^3/s, $P_{\rm phot} = 10^{-4}$ s^{-1}). This recovers the order of magnitude of the C7 tabulated $n_e/n_{\rm H}$ but does *not* match it quantitatively. Model C7 (Avrett and Loeser, 2008) uses full NLTE radiative transfer with multi-level hydrogen and metals; our two-state effective-rate equilibrium misses the height- and density-dependence of the radiative coupling, Balmer/Lyman line scattering, the metal contribution to n_e, and detailed chromospheric radiative heating. This is the same regime in which Leenaarts et al. (2007) show that non-equilibrium ionization rate equations must be carried in time-dependent simulations rather than substituting Saha or any kinetic-equilibrium algebraic relation. The §7 comparison is therefore a regression test, not a validation against C7.

5.6 Scenario-Specific Calibration: C7 Photoionization Profile (Numerical Closure, Not a Physical Rate)

Important caveat read first. This subsection documents a numerical convenience used by the `model_c7` scenario, *not* a physical photoionization rate. The values reported here are obtained by inverting the Stage E fixed-point condition on C7's tabulated (n_e, n_n, T) profile, and inherit every limitation of the 2-state (ground + continuum) reduction used in this code. A faithful physical rate would require the radiative-transfer machinery of Leenaarts et al. (2007) or similar. The calibration described below is the minimum effort to make Model C7 a Stage E fixed point at $t = 0$, so that the chromospheric atmosphere relaxes from a pre-equilibrated state rather than from the recombination-dominated state that a single uniform $P_{\rm phot} \sim 10^{-4}$ s^{-1} produces in the lower chromosphere.

Derivation. For each interior cell i, evaluate the algebraic fixed point of the Stage E ODE (Eq. (101)) on C7's tabulated values $(n_e^{C7}, n_n^{C7}, T^{C7})$ at the cell:

$$P_{\rm phot}^{C7}(s) = \frac{\alpha_r(T^{C7})\,(n_e^{C7})^2}{n_n^{C7}} - S_i(T^{C7})\,n_e^{C7}, \tag{131}$$

with α_r and S_i from Eqs. (129) and (128). The result is non-negative on the C7 profile (round-off-induced negatives are clamped to zero). It is then stored in the cell-centered field `Grid::photoionization_rate_i`; the helper `physics.hpp::photoionization_rate_P` returns this per-cell array in preference to the scalar fallback `Grid::photoionization_rate`.

Table 1 lists the calibrated rate at the fifteen C7 reference heights.

Numerical cap. Without a cap, the implementation crashes around $t \approx 100$ s. The uncapped top-of-domain rate ($P_{\rm phot}^{C7} \sim 2.5 \times 10^{-2}$ s^{-1}) is calibrated to C7's column densities, but under our equations Model C7 is not in hydrostatic equilibrium (writeup §5.3 documents the $\sim 25\%$ imbalance), so bulk flow develops on an acoustic-crossing time of order 100 s. Within that time n_n at the top falls by an order of magnitude while $P_{\rm phot}^{C7}$ stays frozen at its IC value. The photoionization source $P_{\rm phot}^{C7} \cdot n_n$ then dominates the recombination sink $\alpha_r n_e^2$, $f \to 1$ in a few timesteps, and the integrator NaNs. We therefore cap the calibrated rate at $P_{\max} = 10^{-2}$ s^{-1}, the upper bound of the Carlsson and Stein (2002) chromospheric-timescale range. The cap engages in roughly the upper third of the domain ($h \gtrsim 1500$ km); below that, the code uses Eq. (131) verbatim.

h (km)	T_{C7} (K)	n_e^{C7} (m^{-3})	n_n^{C7} (m^{-3})	f_{C7}	P_{phot}^{C7} (s^{-1})
1003	6225	1.90×10^{17}	2.69×10^{19}	0.0070	5.2×10^{-4}
1032	6315	2.02×10^{17}	2.18×10^{19}	0.0092	7.1×10^{-4}
1065	6400	2.09×10^{17}	1.72×10^{19}	0.0120	9.6×10^{-4}
1101	6474	2.10×10^{17}	1.34×10^{19}	0.0155	1.2×10^{-3}
1143	6531	2.00×10^{17}	1.01×10^{19}	0.0194	1.5×10^{-3}
1214	6576	1.76×10^{17}	6.32×10^{18}	0.0271	1.8×10^{-3}
1299	6598	1.49×10^{17}	3.64×10^{18}	0.0393	2.2×10^{-3}
1398	6610	1.44×10^{17}	1.88×10^{18}	0.0711	4.1×10^{-3}
1520	6623	1.42×10^{17}	7.96×10^{17}	0.1517	9.4×10^{-3}
1617	6633	1.27×10^{17}	4.01×10^{17}	0.2400	1.5×10^{-2}
1722	6643	1.03×10^{17}	1.92×10^{17}	0.3490	2.0×10^{-2}
1820	6652	7.97×10^{16}	9.84×10^{16}	0.4474	2.4×10^{-2}
1894	6660	6.45×10^{16}	6.01×10^{16}	0.5179	2.5×10^{-2}
1946	6667	5.53×10^{16}	4.37×10^{16}	0.5584	2.6×10^{-2}
1989	6674	4.83×10^{16}	3.45×10^{16}	0.5833	2.5×10^{-2}

Table 1: C7-calibrated photoionization rate P_{phot}^{C7} obtained by inverting Eq. (131) on the Avrett and Loeser (2008) tabulated atmosphere. These are the raw values before the stability cap $P_{\max} = 10^{-2}\ \mathrm{s}^{-1}$ (see "Numerical cap" below). The code interpolates this profile onto every grid cell after spline interpolation of $(n_e^{C7}, n_n^{C7}, T^{C7})$.

Three reasons this is not a physical rate.

1. *The inversion absorbs every gap between C7 and our model into a single number.* Eq. (131) returns whatever value our 2-state Stage E equation *requires* to reproduce C7's tabulated f, given the chosen α_r and S_i. If our model misses physics that C7 includes, the inversion absorbs the missing physics into P_{phot}^{C7} rather than identifying it.

2. *Inflated by metal contributions to n_e.* Avrett and Loeser (2008) include Ca, Mg, Fe, Si, Al, Na, K, and S — each at low first-ionization potential, each a significant electron donor at chromospheric temperatures. Roughly 30–50% of n_e in the mid-chromosphere comes from metals, not from ionized hydrogen. Our calibration attributes *all* of n_e to ionized H, so the back-solved P_{phot}^{C7} is too large by a factor of a few in the mid-chromosphere.

3. *Stops being applicable as soon as bulk flow begins.* Eq. (131) uses C7's (n_e, n_n, T), but $P_{\mathrm{phot}}^{C7}(s)$ is stored at the IC and stays fixed while the simulated densities evolve. The calibration is exact only at $t = 0^+$; the more the simulation drifts from C7 (which it does everywhere except the lower $\sim 30\%$ of the domain), the further the rate is from the rate that would make the *current* state a fixed point.

Comparison to verifiable rates. The Carlsson & Stein 2002 timescale range $\tau_{\mathrm{ion}} \in [10^3, 10^5]$ s gives $P_{\mathrm{phot}} \in [10^{-5}, 10^{-3}]\ \mathrm{s}^{-1}$. Table 1 compared against this range:

- Lower chromosphere ($h \in [1003, 1200]$ km): $P_{\mathrm{phot}}^{C7} \in [5 \times 10^{-4},\ 2 \times 10^{-3}]\ \mathrm{s}^{-1}$, consistent with the upper end of the C&S 2002 range. *Physically defensible.*

- Mid-chromosphere ($h \in [1200, 1500]$ km): P_{phot}^{C7} reaches $\sim 10^{-2}\ \mathrm{s}^{-1}$, an order of magnitude above the C&S 2002 quiescent range. *Inflated, most likely by the metal-n_e contamination noted above.*

- Upper chromosphere ($h \in [1500, 1989]$ km): uncapped P_{phot}^{C7} ranges 1.5–$2.6 \times 10^{-2}\ \mathrm{s}^{-1}$, consistent only with the *shock-heated extreme* of Carlsson and Stein (2002), not with the quiet mean. Capped at $10^{-2}\ \mathrm{s}^{-1}$ for stability; the cap aligns with the top of the verifiable range.

The simulation outcome (writeup §7 comparison plots) reflects this gradient: $f_{\mathrm{sim}}/f_{C7} \sim 1$ in the lower chromosphere where the calibration is both physically defensible and numerically applicable, and grows to ~ 2.5 in the middle where the metal-contamination and the cap dominate.

Code interface. The calibration is performed at scenario IC time in `scenarios/model_c7.cpp::model_c7_ic` and is specific to the `model_c7` scenario. Other scenarios (`analytic_canopy`, `pfss_field_line`) leave `Grid::photoionization_rate_i` empty and use the uniform scalar default $P_{\rm phot} = 10^{-4}$ s^{-1} (`Grid::photoionization_rate`) from §5.5. Replacing the calibrated profile with a height-dependent radiative-transfer-derived profile drops in by overwriting the body of `model_c7_ic`'s `photoionization_rate_i` block.

6 Numerical Results

The initial conditions are defined by Model C7 (Avrett and Loeser, 2008), as listed in Table 2. A straight field line of 100 equal-length cells is generated by spline interpolation. The magnetic field B is held constant at 1 T, and gravitational forces are not yet included.

For boundary conditions, the inner boundary is held fixed at the initial state. At the outer boundary, the outermost-ghost temperature is set to twice that of the outermost interior cell, and the velocity is updated each step to half of the outermost-cell value.

Height (km)	**Electron Density (m^{-3})**	**H Atom Density (m^{-3})**	**Temperature (K)**
1.003e+03	1.903e+17	2.693e+19	6.225e+03
1.032e+03	2.021e+17	2.179e+19	6.315e+03
1.065e+03	2.091e+17	1.722e+19	6.400e+03
1.101e+03	2.104e+17	1.340e+19	6.474e+03
1.143e+03	1.995e+17	1.009e+19	6.531e+03
1.214e+03	1.760e+17	6.322e+18	6.576e+03
1.299e+03	1.489e+17	3.641e+18	6.598e+03
1.398e+03	1.438e+17	1.880e+18	6.610e+03
1.520e+03	1.423e+17	7.956e+17	6.623e+03
1.617e+03	1.267e+17	4.012e+17	6.633e+03
1.722e+03	1.027e+17	1.916e+17	6.643e+03
1.820e+03	7.969e+16	9.843e+16	6.652e+03
1.894e+03	6.450e+16	6.005e+16	6.660e+03
1.946e+03	5.526e+16	4.371e+16	6.667e+03
1.989e+03	4.826e+16	3.447e+16	6.674e+03

Table 2: Height, Electron Number Density, H Atom Number Density, and Temperature.

The ion sound speed C_{si} for this profile is approximately 1.3×10^4 m/s. To reach a steady-state solution, the total simulation time is set to $100\,L/C_{si}$, where L is the length of the field line, i.e. 0.986×10^3 km.

Ionization fraction. With Stage E (§4.4) enabled, the solver evolves the ionization fraction $f(s,t) = \rho_i/(\rho_i+\rho_n)$ self-consistently from the initial Model C7 profile through the combined action of advection, conduction, drag, and the ionization–recombination rate equations of §2.3. The expected steady state at each cell is the kinetic equilibrium $f_{\rm eq}/(1 - f_{\rm eq}) = S_i(T_e)/\alpha_r(T_e)$ with T_e self-consistently shifted by the χ_H drain on the electron thermal pool. As discussed in §5.5, this will agree with Model C7's tabulated $n_e/n_{\rm H}$ only at the order-of-magnitude level: the local rate balance does not include NLTE line scattering, metal ionization, or chromospheric radiative heating. A panel comparing the computed $f(s)$ to Model C7's tabulated ionization fraction will be added in a follow-up revision; it serves as a regression diagnostic — confirming that the rate equations are stably integrated, that $\rho_i + \rho_n$ is conserved to round-off, and that the optically-thin radiative loss in Eq. (15) acts as the expected energy sink — rather than a validation against Avrett and Loeser (2008). A regime in which the χ_H drain is balanced by a chromospheric heating term is deferred to a future extension of the model.

A Multi-Dimensional Numerical Framework

We collect the 14 unknowns of the normalized two-fluid + GLM system into a single state vector and assemble the corresponding x- and z-direction fluxes. The flux Jacobians and their eigen systems determine the wave speeds used to construct the numerical fluxes. The ionization–recombination source terms of §2.3 are pure body sources with no divergence form, so they enter only on the right-hand side and *do not* alter the flux vector, the flux Jacobians, or the eigen system derived in this section. Their numerical handling is deferred to §4.4 as an additional stiff block of the implicit residual $\mathbf{R}_{I,i}$.

A.1 State Vector and Direction Fluxes

$$\mathbf{u} = \begin{pmatrix} n_\mathrm{i} \\ n_\mathrm{n} \\ n_\mathrm{i} v_\mathrm{x} \\ n_\mathrm{i} v_\mathrm{y} \\ n_\mathrm{i} v_\mathrm{z} \\ n_\mathrm{n} u_\mathrm{x} \\ n_\mathrm{n} u_\mathrm{y} \\ n_\mathrm{n} u_\mathrm{z} \\ b_\mathrm{x} \\ b_\mathrm{y} \\ b_\mathrm{z} \\ T_\mathrm{i} \\ T_\mathrm{n} \\ \Phi \end{pmatrix}, \mathbf{F}_\mathrm{x} = \begin{pmatrix} n_\mathrm{i} v_\mathrm{x} \\ n_\mathrm{n} u_\mathrm{x} \\ \frac{b_2}{2} - b_\mathrm{x}^2 + n_\mathrm{i} T_\mathrm{i} + n_\mathrm{i} v_\mathrm{x}^2 \\ -b_\mathrm{x} b_\mathrm{y} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{y} \\ -b_\mathrm{x} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{z} \\ \frac{n_\mathrm{n} T_\mathrm{n}}{m_\mathrm{n}} + n_\mathrm{n} u_\mathrm{x}^2 \\ n_\mathrm{n} u_\mathrm{x} u_\mathrm{y} \\ n_\mathrm{n} u_\mathrm{x} u_\mathrm{z} \\ -\Phi \\ b_\mathrm{y} v_\mathrm{x} - b_\mathrm{x} v_\mathrm{y} \\ b_\mathrm{z} v_\mathrm{x} - b_\mathrm{x} v_\mathrm{z} \\ T_\mathrm{i} v_\mathrm{x} \\ T_\mathrm{n} u_\mathrm{x} \\ b_\mathrm{x} c_\mathrm{h}^2 \end{pmatrix}, \mathbf{F}_\mathrm{z} = \begin{pmatrix} n_\mathrm{i} v_\mathrm{z} \\ n_\mathrm{n} u_\mathrm{z} \\ -b_\mathrm{x} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{z} \\ -b_\mathrm{y} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{y} v_\mathrm{z} \\ \frac{b_2}{2} - b_\mathrm{z}^2 + n_\mathrm{i} T_\mathrm{i} + n_\mathrm{i} v_\mathrm{z}^2 \\ n_\mathrm{n} u_\mathrm{x} u_\mathrm{z} \\ n_\mathrm{n} u_\mathrm{y} u_\mathrm{z} \\ \frac{n_\mathrm{n} T_\mathrm{n}}{m_\mathrm{n}} + n_\mathrm{n} u_\mathrm{z}^2 \\ -b_\mathrm{z} v_\mathrm{x} + b_\mathrm{x} v_\mathrm{z} \\ -b_\mathrm{z} v_\mathrm{y} + b_\mathrm{y} v_\mathrm{z} \\ -\Phi \\ T_\mathrm{i} v_\mathrm{z} \\ T_\mathrm{n} u_\mathrm{z} \\ b_\mathrm{z} c_\mathrm{h}^2 \end{pmatrix} \tag{132}$$

For reference, the individual block fluxes used to assemble $\mathbf{F_x}$ are

$$F_\mathrm{ni} = \begin{pmatrix} n_\mathrm{i}\, v_\mathrm{x} \\ n_\mathrm{i}\, v_\mathrm{y} \\ n_\mathrm{i}\, v_\mathrm{z} \end{pmatrix}, \quad F_\mathrm{nn} = \begin{pmatrix} n_\mathrm{n}\, u_\mathrm{x} \\ n_\mathrm{n}\, u_\mathrm{y} \\ n_\mathrm{n}\, u_\mathrm{z} \end{pmatrix}, \quad F_\mathrm{Ti} = \begin{pmatrix} T_\mathrm{i} v_\mathrm{x} \\ T_\mathrm{i} v_\mathrm{y} \\ T_\mathrm{i} v_\mathrm{z} \end{pmatrix}, \quad F_\mathrm{Tn} = \begin{pmatrix} T_\mathrm{n} u_\mathrm{x} \\ T_\mathrm{n} u_\mathrm{y} \\ T_\mathrm{n} u_\mathrm{z} \end{pmatrix},$$

$$F_\Phi = \begin{pmatrix} c_\mathrm{h}^2 b_\mathrm{x} \\ c_\mathrm{h}^2 b_\mathrm{y} \\ c_\mathrm{h}^2 b_\mathrm{z} \end{pmatrix}, \quad F_\mathrm{V} = \begin{pmatrix} \frac{b^2}{2} - b_\mathrm{x}^2 + n_\mathrm{i} T_\mathrm{i} + n_\mathrm{i} v_\mathrm{x}^2 & -b_\mathrm{x} b_\mathrm{y} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{y} & -b_\mathrm{x} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{z} \\ -b_\mathrm{x} b_\mathrm{y} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{y} & \frac{b^2}{2} - b_\mathrm{y}^2 + n_\mathrm{i} T_\mathrm{i} + n_\mathrm{i} v_\mathrm{y}^2 & -b_\mathrm{y} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{y} v_\mathrm{z} \\ -b_\mathrm{x} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{x} v_\mathrm{z} & -b_\mathrm{y} b_\mathrm{z} + n_\mathrm{i} v_\mathrm{y} v_\mathrm{z} & \frac{b^2}{2} - b_\mathrm{z}^2 + n_\mathrm{i} T_\mathrm{i} + n_\mathrm{i} v_\mathrm{z}^2 \end{pmatrix},$$

$$F_\mathrm{U} = \begin{pmatrix} \frac{n_\mathrm{n} T_\mathrm{n}}{m_\mathrm{n}} + n_\mathrm{n} u_\mathrm{x}^2 & n_\mathrm{n} u_\mathrm{x} u_\mathrm{y} & n_\mathrm{n} u_\mathrm{x} u_\mathrm{z} \\ n_\mathrm{n} u_\mathrm{x} u_\mathrm{y} & \frac{n_\mathrm{n} T_\mathrm{n}}{m_\mathrm{n}} + n_\mathrm{n} u_\mathrm{y}^2 & n_\mathrm{n} u_\mathrm{y} u_\mathrm{z} \\ n_\mathrm{n} u_\mathrm{x} u_\mathrm{z} & n_\mathrm{n} u_\mathrm{y} u_\mathrm{z} & \frac{n_\mathrm{n} T_\mathrm{n}}{m_\mathrm{n}} + n_\mathrm{n} u_\mathrm{z}^2 \end{pmatrix}, \quad F_\mathrm{B} = \begin{pmatrix} -\Phi & b_\mathrm{y} v_\mathrm{x} - b_\mathrm{x} v_\mathrm{y} & b_\mathrm{z} v_\mathrm{x} - b_\mathrm{x} v_\mathrm{z} \\ -b_\mathrm{y} v_\mathrm{x} + b_\mathrm{x} v_\mathrm{y} & -\Phi & b_\mathrm{z} v_\mathrm{y} - b_\mathrm{y} v_\mathrm{z} \\ -b_\mathrm{z} v_\mathrm{x} + b_\mathrm{x} v_\mathrm{z} & -b_\mathrm{z} v_\mathrm{y} + b_\mathrm{y} v_\mathrm{z} & -\Phi \end{pmatrix}.$$

The corresponding velocity-derivative blocks $G = \frac{\partial}{\partial u_j} F_{ij}$ used in upwinding are

$$G_\mathrm{V} = \begin{pmatrix} 2\, n_\mathrm{i}\, v_\mathrm{x} & n_\mathrm{i}\, v_\mathrm{x} & n_\mathrm{i}\, v_\mathrm{x} \\ n_\mathrm{i}\, v_\mathrm{y} & 2\, n_\mathrm{i}\, v_\mathrm{y} & n_\mathrm{i}\, v_\mathrm{y} \\ n_\mathrm{i}\, v_\mathrm{z} & n_\mathrm{i}\, v_\mathrm{z} & 2\, n_\mathrm{i}\, v_\mathrm{z} \end{pmatrix}, \quad G_\mathrm{U} = \begin{pmatrix} 2\, n_\mathrm{n}\, u_\mathrm{x} & n_\mathrm{n}\, u_\mathrm{x} & n_\mathrm{n}\, u_\mathrm{x} \\ n_\mathrm{n}\, u_\mathrm{y} & 2\, n_\mathrm{n}\, u_\mathrm{y} & n_\mathrm{n}\, u_\mathrm{y} \\ n_\mathrm{n}\, u_\mathrm{z} & n_\mathrm{n}\, u_\mathrm{z} & 2\, n_\mathrm{n}\, u_\mathrm{z} \end{pmatrix},$$

$$G_\mathrm{B} = \begin{pmatrix} 0 & v_\mathrm{x} & v_\mathrm{x} \\ v_\mathrm{y} & 0 & v_\mathrm{y} \\ v_\mathrm{z} & v_\mathrm{z} & 0 \end{pmatrix}, \quad G_{ni} = \begin{pmatrix} v_\mathrm{x} \\ v_\mathrm{y} \\ v_\mathrm{z} \end{pmatrix}, \quad G_{nn} = \begin{pmatrix} u_\mathrm{x} \\ u_\mathrm{y} \\ u_\mathrm{z} \end{pmatrix}, \quad G_{Ti} = \begin{pmatrix} v_\mathrm{x} \\ v_\mathrm{y} \\ v_\mathrm{z} \end{pmatrix}, \quad G_{Tn} = \begin{pmatrix} u_\mathrm{x} \\ u_\mathrm{y} \\ u_\mathrm{z} \end{pmatrix}.$$

A.2 Flux Jacobians

$$
\frac{\mathrm{d}\mathbf{F_x}}{\mathrm{d}\mathbf{u}} = \begin{pmatrix}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
T_\mathrm{i} - v_\mathrm{x}^2 & 0 & 2v_\mathrm{x} & 0 & 0 & 0 & 0 & 0 & -b_\mathrm{x} & b_\mathrm{y} & b_\mathrm{z} & n_\mathrm{i} & 0 & 0 \\
-v_\mathrm{x}v_\mathrm{y} & 0 & v_\mathrm{y} & v_\mathrm{x} & 0 & 0 & 0 & 0 & -b_\mathrm{y} & -b_\mathrm{x} & 0 & 0 & 0 & 0 \\
-v_\mathrm{x}v_\mathrm{z} & 0 & v_\mathrm{z} & 0 & v_\mathrm{x} & 0 & 0 & 0 & -b_\mathrm{z} & 0 & -b_\mathrm{x} & 0 & 0 & 0 \\
0 & \frac{T_\mathrm{n}}{m_\mathrm{n}} - u_\mathrm{x}^2 & 0 & 0 & 0 & 2u_\mathrm{x} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{n_\mathrm{n}}{m_\mathrm{n}} & 0 \\
0 & -u_\mathrm{x}u_\mathrm{y} & 0 & 0 & 0 & u_\mathrm{y} & u_\mathrm{x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -u_\mathrm{x}u_\mathrm{z} & 0 & 0 & 0 & u_\mathrm{z} & 0 & u_\mathrm{x} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
\frac{-b_\mathrm{y}v_\mathrm{x}+b_\mathrm{x}v_\mathrm{y}}{n_\mathrm{i}} & 0 & \frac{b_\mathrm{y}}{n_\mathrm{i}} & -\frac{b_\mathrm{x}}{n_\mathrm{i}} & 0 & 0 & 0 & 0 & -v_\mathrm{y} & v_\mathrm{x} & 0 & 0 & 0 & 0 \\
\frac{-b_\mathrm{z}v_\mathrm{x}+b_\mathrm{x}v_\mathrm{z}}{n_\mathrm{i}} & 0 & \frac{b_\mathrm{z}}{n_\mathrm{i}} & 0 & -\frac{b_\mathrm{x}}{n_\mathrm{i}} & 0 & 0 & 0 & -v_\mathrm{z} & 0 & v_\mathrm{x} & 0 & 0 & 0 \\
-\frac{T_\mathrm{i}v_\mathrm{x}}{n_\mathrm{i}} & 0 & \frac{T_\mathrm{i}}{n_\mathrm{i}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & v_\mathrm{x} & 0 & 0 \\
0 & -\frac{T_\mathrm{n}u_\mathrm{x}}{n_\mathrm{n}} & 0 & 0 & 0 & \frac{T_\mathrm{n}}{n_\mathrm{n}} & 0 & 0 & 0 & 0 & 0 & 0 & u_\mathrm{x} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_\mathrm{h}^2 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}, \tag{133}
$$

$$
\frac{\mathrm{d}\mathbf{F_z}}{\mathrm{d}\mathbf{u}} = \begin{pmatrix}
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
-v_\mathrm{x}v_\mathrm{z} & 0 & v_\mathrm{z} & 0 & v_\mathrm{x} & 0 & 0 & 0 & -b_\mathrm{z} & 0 & -b_\mathrm{x} & 0 & 0 & 0 \\
-v_\mathrm{y}v_\mathrm{z} & 0 & 0 & v_\mathrm{z} & v_\mathrm{y} & 0 & 0 & 0 & 0 & -b_\mathrm{z} & -b_\mathrm{y} & 0 & 0 & 0 \\
T_\mathrm{i} - v_\mathrm{z}^2 & 0 & 0 & 0 & 2v_\mathrm{z} & 0 & 0 & 0 & b_\mathrm{x} & b_\mathrm{y} & -b_\mathrm{z} & n_\mathrm{i} & 0 & 0 \\
0 & -u_\mathrm{x}u_\mathrm{z} & 0 & 0 & 0 & u_\mathrm{z} & 0 & u_\mathrm{x} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -u_\mathrm{y}u_\mathrm{z} & 0 & 0 & 0 & 0 & u_\mathrm{z} & u_\mathrm{y} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{T_\mathrm{n}}{m_\mathrm{n}} - u_\mathrm{z}^2 & 0 & 0 & 0 & 0 & 0 & 2u_\mathrm{z} & 0 & 0 & 0 & 0 & \frac{n_\mathrm{n}}{m_\mathrm{n}} & 0 \\
\frac{b_\mathrm{z}v_\mathrm{x}-b_\mathrm{x}v_\mathrm{z}}{n_\mathrm{i}} & 0 & -\frac{b_\mathrm{z}}{n_\mathrm{i}} & 0 & \frac{b_\mathrm{x}}{n_\mathrm{i}} & 0 & 0 & 0 & v_\mathrm{z} & 0 & -v_\mathrm{x} & 0 & 0 & 0 \\
\frac{b_\mathrm{z}v_\mathrm{y}-b_\mathrm{y}v_\mathrm{z}}{n_\mathrm{i}} & 0 & 0 & -\frac{b_\mathrm{z}}{n_\mathrm{i}} & \frac{b_\mathrm{y}}{n_\mathrm{i}} & 0 & 0 & 0 & 0 & v_\mathrm{z} & -v_\mathrm{y} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
-\frac{T_\mathrm{i}v_\mathrm{z}}{n_\mathrm{i}} & 0 & 0 & 0 & \frac{T_\mathrm{i}}{n_\mathrm{i}} & 0 & 0 & 0 & 0 & 0 & 0 & v_\mathrm{z} & 0 & 0 \\
0 & -\frac{T_\mathrm{n}u_\mathrm{z}}{n_\mathrm{n}} & 0 & 0 & 0 & 0 & 0 & \frac{T_\mathrm{n}}{n_\mathrm{n}} & 0 & 0 & 0 & 0 & u_\mathrm{z} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_\mathrm{h}^2 & 0 & 0 & 0 & 0 & 0
\end{pmatrix} \tag{134}
$$

A.3 Eigenvalues and Fast-Magnetosonic Speeds

The eigenvalues are

$$
\lambda\left(\frac{\mathrm{d}\mathbf{F_x}}{\mathrm{d}\mathbf{u}}\right) = \begin{pmatrix}
-ic_\mathrm{h} \\
ic_\mathrm{h} \\
u_\mathrm{x} \\
u_\mathrm{x} \\
u_\mathrm{x} \\
-\sqrt{2}\sqrt{\frac{T_\mathrm{n}}{m_\mathrm{n}}} + u_\mathrm{x} \\
\sqrt{2}\sqrt{\frac{T_\mathrm{n}}{m_\mathrm{n}}} + u_\mathrm{x} \\
v_\mathrm{x} \\
-\frac{b_\mathrm{x}}{\sqrt{n_\mathrm{i}}} + v_\mathrm{x} \\
\frac{b_\mathrm{x}}{\sqrt{n_\mathrm{i}}} + v_\mathrm{x} \\
-\frac{\sqrt{n_\mathrm{i}\left(b_\mathrm{x}^2+b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i}-\sqrt{b_\mathrm{x}^4+2b_\mathrm{x}^2(b_\mathrm{y}^2+b_\mathrm{z}^2-2n_\mathrm{i}T_\mathrm{i})+(b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i})^2}\right)}}{\sqrt{2}n_\mathrm{i}} + v_\mathrm{x} \\
\frac{\sqrt{n_\mathrm{i}\left(b_\mathrm{x}^2+b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i}-\sqrt{b_\mathrm{x}^4+2b_\mathrm{x}^2(b_\mathrm{y}^2+b_\mathrm{z}^2-2n_\mathrm{i}T_\mathrm{i})+(b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i})^2}\right)}}{\sqrt{2}n_\mathrm{i}} + v_\mathrm{x} \\
-\frac{\sqrt{n_\mathrm{i}\left(b_\mathrm{x}^2+b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i}+\sqrt{b_\mathrm{x}^4+2b_\mathrm{x}^2(b_\mathrm{y}^2+b_\mathrm{z}^2-2n_\mathrm{i}T_\mathrm{i})+(b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i})^2}\right)}}{\sqrt{2}n_\mathrm{i}} + v_\mathrm{x} \\
\frac{\sqrt{n_\mathrm{i}\left(b_\mathrm{x}^2+b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i}+\sqrt{b_\mathrm{x}^4+2b_\mathrm{x}^2(b_\mathrm{y}^2+b_\mathrm{z}^2-2n_\mathrm{i}T_\mathrm{i})+(b_\mathrm{y}^2+b_\mathrm{z}^2+2n_\mathrm{i}T_\mathrm{i})^2}\right)}}{\sqrt{2}n_\mathrm{i}} + v_\mathrm{x}
\end{pmatrix}, \tag{135}
$$

$$
\lambda(\frac{\mathrm{d}\mathbf{F_z}}{\mathrm{d}\mathbf{u}}) = \begin{pmatrix}
-ic_{\mathrm{h}} \\
ic_{\mathrm{h}} \\
u_{\mathrm{z}} \\
u_{\mathrm{z}} \\
u_{\mathrm{z}} \\
-\sqrt{2}\sqrt{\frac{T_{\mathrm{n}}}{m_{\mathrm{n}}}} + u_{\mathrm{z}} \\
\sqrt{2}\sqrt{\frac{T_{\mathrm{n}}}{m_{\mathrm{n}}}} + u_{\mathrm{z}} \\
v_{\mathrm{z}} \\
-\frac{b_{\mathrm{z}}}{\sqrt{n_{\mathrm{i}}}} + v_{\mathrm{z}} \\
\frac{b_{\mathrm{z}}}{\sqrt{n_{\mathrm{i}}}} + v_{\mathrm{z}} \\
-\frac{\sqrt{n_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+2n_{\mathrm{i}}T_{\mathrm{i}}-\sqrt{b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+4n_{\mathrm{i}}T_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2-b_{\mathrm{z}}^2\right)+4n_{\mathrm{i}}^2T_{\mathrm{i}}^2}\right)}}{\sqrt{2}n_{\mathrm{i}}} + v_{\mathrm{z}} \\
\frac{\sqrt{n_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+2n_{\mathrm{i}}T_{\mathrm{i}}-\sqrt{b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+4n_{\mathrm{i}}T_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2-b_{\mathrm{z}}^2\right)+4n_{\mathrm{i}}^2T_{\mathrm{i}}^2}\right)}}{\sqrt{2}n_{\mathrm{i}}} + v_{\mathrm{z}} \\
-\frac{\sqrt{n_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+2n_{\mathrm{i}}T_{\mathrm{i}}+\sqrt{b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+4n_{\mathrm{i}}T_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2-b_{\mathrm{z}}^2\right)+4n_{\mathrm{i}}^2T_{\mathrm{i}}^2}\right)}}{\sqrt{2}n_{\mathrm{i}}} + v_{\mathrm{z}} \\
\frac{\sqrt{n_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+2n_{\mathrm{i}}T_{\mathrm{i}}+\sqrt{b_{\mathrm{x}}^2+b_{\mathrm{y}}^2+b_{\mathrm{z}}^2+4n_{\mathrm{i}}T_{\mathrm{i}}\left(b_{\mathrm{x}}^2+b_{\mathrm{y}}^2-b_{\mathrm{z}}^2\right)+4n_{\mathrm{i}}^2T_{\mathrm{i}}^2}\right)}}{\sqrt{2}n_{\mathrm{i}}} + v_{\mathrm{z}}
\end{pmatrix}, \tag{136}
$$

which can be simplified as

$$
\lambda(\frac{\mathrm{d}\mathbf{F_x}}{\mathrm{d}\mathbf{u}}) = \begin{pmatrix}
-ic_{\mathrm{h}} \\
ic_{\mathrm{h}} \\
u_{\mathrm{x}} \\
u_{\mathrm{x}} \\
u_{\mathrm{x}} \\
-\sqrt{2}\sqrt{\frac{T_{\mathrm{n}}}{m_{\mathrm{n}}}} + u_{\mathrm{x}} \\
\sqrt{2}\sqrt{\frac{T_{\mathrm{n}}}{m_{\mathrm{n}}}} + u_{\mathrm{x}} \\
v_{\mathrm{x}} \\
-\frac{b_{\mathrm{x}}}{\sqrt{n_{\mathrm{i}}}} + v_{\mathrm{x}} \\
\frac{b_{\mathrm{x}}}{\sqrt{n_{\mathrm{i}}}} + v_{\mathrm{x}} \\
-c_{\mathrm{fast,x}}^{-} + v_{\mathrm{x}} \\
c_{\mathrm{fast,x}}^{-} + v_{\mathrm{x}} \\
-c_{\mathrm{fast,x}}^{+} + v_{\mathrm{x}} \\
c_{\mathrm{fast,x}}^{+} + v_{\mathrm{x}}
\end{pmatrix}, \tag{137}
$$

$$
\lambda(\frac{\mathrm{d}\mathbf{F_z}}{\mathrm{d}\mathbf{u}}) = \begin{pmatrix}
-ic_{\mathrm{h}} \\
ic_{\mathrm{h}} \\
u_{\mathrm{z}} \\
u_{\mathrm{z}} \\
u_{\mathrm{z}} \\
-\sqrt{2}\sqrt{\frac{T_{\mathrm{n}}}{m_{\mathrm{n}}}} + u_{\mathrm{z}} \\
\sqrt{2}\sqrt{\frac{T_{\mathrm{n}}}{m_{\mathrm{n}}}} + u_{\mathrm{z}} \\
v_{\mathrm{z}} \\
-\frac{b_{\mathrm{z}}}{\sqrt{n_{\mathrm{i}}}} + v_{\mathrm{z}} \\
\frac{b_{\mathrm{z}}}{\sqrt{n_{\mathrm{i}}}} + v_{\mathrm{z}} \\
-c_{\mathrm{fast,z}}^{-} + v_{\mathrm{z}} \\
c_{\mathrm{fast,z}}^{-} + v_{\mathrm{z}} \\
-c_{\mathrm{fast,z}}^{+} + v_{\mathrm{z}} \\
c_{\mathrm{fast,z}}^{+} + v_{\mathrm{z}}
\end{pmatrix}, \tag{138}
$$

where

$$c^{+}_{\mathrm{fast,x}} = \frac{1}{\sqrt{2}}\sqrt{c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2 + \sqrt{(c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2)^2 - \frac{8}{\gamma}c_{\mathrm{A,x}}^2 c_{\mathrm{s}}^2}}, \tag{139}$$

$$c^{+}_{\mathrm{fast,z}} = \frac{1}{\sqrt{2}}\sqrt{c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2 + \sqrt{(c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2)^2 - \frac{8}{\gamma}c_{\mathrm{A,z}}^2 c_{\mathrm{s}}^2}}, \tag{140}$$

$$c^{-}_{\mathrm{fast,x}} = \frac{1}{\sqrt{2}}\sqrt{c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2 - \sqrt{(c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2)^2 - \frac{8}{\gamma}c_{\mathrm{A,x}}^2 c_{\mathrm{s}}^2}}, \tag{141}$$

$$c^{-}_{\mathrm{fast,z}} = \frac{1}{\sqrt{2}}\sqrt{c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2 - \sqrt{(c_{\mathrm{A}}^2 + \frac{2}{\gamma}c_{\mathrm{s}}^2)^2 - \frac{8}{\gamma}c_{\mathrm{A,z}}^2 c_{\mathrm{s}}^2}}, \tag{142}$$

and

$$c_{\mathrm{A}} = \sqrt{\frac{b_{\mathrm{x}}^2 + b_{\mathrm{y}}^2 + b_{\mathrm{z}}^2}{n_{\mathrm{i}}}}, c_{\mathrm{A,x}} = \frac{b_{\mathrm{x}}}{\sqrt{n_{\mathrm{i}}}}, \quad c_{\mathrm{A,z}} = \frac{b_{\mathrm{z}}}{\sqrt{n_{\mathrm{i}}}}, \quad c_{\mathrm{s}} = \sqrt{\gamma T_{\mathrm{i}}}. \tag{143}$$

This completes the abstract framework: state vector, flux, Jacobian, and eigen system. The corresponding spatial and temporal discretizations (TVD–MUSCL reconstruction, Rusanov flux, predictor–corrector, and semi-implicit time integration) are deferred to Section 4, after the field-aligned reduction of Section 3 renders the conservative form and the explicit/implicit residual partition concrete.

References

Qusai Al Shidi, Ofer Cohen, Paul Song, and Jiannan Tu. Time-dependent two-fluid magnetohydrodynamic model and simulation of the chromosphere. *Solar Physics*, 294(9):124, 2019. doi: 10.1007/s11207-019-1513-8.

U. M. Ascher, S. J. Ruuth, and R. J. Spiteri. Implicit-explicit Runge-Kutta methods for time-dependent partial differential equations. *Applied Numerical Mathematics*, 25:151–167, 1997. doi: 10.1016/S0168-9274(97)00056-1.

E. H. Avrett and R. Loeser. Models of the solar chromosphere and transition region from SUMER and HRTS observations: Formation of the extreme-ultraviolet spectrum of hydrogen, carbon, and oxygen. *The Astrophysical Journal Supplement Series*, 175:229–276, 2008. doi: 10.1086/523671.

M. Carlsson and J. Leenaarts. Approximations for radiative cooling and heating in the solar chromosphere. *Astronomy & Astrophysics*, 539:A39, 2012. doi: 10.1051/0004-6361/201118366.

M. Carlsson and R. F. Stein. Dynamic hydrogen ionization. *The Astrophysical Journal*, 572:626–635, 2002. doi: 10.1086/340293.

A. Dedner, F. Kemm, D. Kröner, C.-D. Munz, T. Schnitzer, and M. Wesenberg. Hyperbolic divergence cleaning for the MHD equations. *Journal of Computational Physics*, 175:645–673, 2002. doi: 10.1006/jcph.2001.6961.

D. G. Hummer. Total recombination and energy loss coefficients for hydrogenic ions at low density for $10 \leq T_{\mathrm{e}}/Z^2 \leq 10^7$ K. *Monthly Notices of the Royal Astronomical Society*, 268:109–112, 1994. doi: 10.1093/mnras/268.1.109.

E. Khomenko, M. Collados, A. Díaz, and N. Vitas. Fluid description of multi-component solar partially ionized plasma. *Physics of Plasmas*, 21(9):092901, 2014. doi: 10.1063/1.4894106.

J. E. Leake, V. S. Lukin, M. G. Linton, and E. T. Meier. Multi-fluid simulations of chromospheric magnetic reconnection in a weakly ionized reacting plasma. *The Astrophysical Journal*, 760:109, 2012. doi: 10.1088/0004-637X/760/2/109.

J. Leenaarts, M. Carlsson, V. Hansteen, and R. J. Rutten. Non-equilibrium hydrogen ionization in 2D simulations of the solar atmosphere. *Astronomy & Astrophysics*, 473:625–632, 2007. doi: 10.1051/0004-6361:20078161.

Y. G. Maneva, A. Alvarez Laguna, A. Lani, and S. Poedts. Multi-fluid modeling of magnetosonic wave propagation in the solar chromosphere: Effects of impact ionization and radiative recombination. *The Astrophysical Journal*, 836:197, 2017. doi: 10.3847/1538-4357/aa5b83.

E. T. Meier and U. Shumlak. A general nonlinear fluid model for reacting plasma–neutral mixtures. *Physics of Plasmas*, 19(7):072508, 2012. doi: 10.1063/1.4736975.

L. Pareschi and G. Russo. Implicit-explicit Runge-Kutta schemes and applications to hyperbolic systems with relaxation. *Journal of Scientific Computing*, 25:129–155, 2005. doi: 10.1007/s10915-004-4636-4.

B. Popescu Braileanu and R. Keppens. Two-fluid implementation in MPI-AMRVAC with applications in the solar chromosphere. *Astronomy & Astrophysics*, 664:A55, 2022. doi: 10.1051/0004-6361/202243630.

B. Popescu Braileanu, V. S. Lukin, E. Khomenko, and Á. de Vicente. Two-fluid simulations of waves in the solar chromosphere. I. numerical code verification. *Astronomy & Astrophysics*, 646:A93, 2021. doi: 10.1051/0004-6361/202039053.

B. Snow and A. Hillier. Collisional ionisation, recombination, and ionisation potential in two-fluid slow-mode shocks: Analytical and numerical results. *Astronomy & Astrophysics*, 637:A97, 2020. doi: 10.1051/0004-6361/201937848.

B. Snow, M. K. Druett, and A. Hillier. Partially-ionised two-fluid shocks with collisional and radiative ionisation and recombination – multi-level hydrogen model. *Monthly Notices of the Royal Astronomical Society*, 525:4717–4734, 2023. doi: 10.1093/mnras/stad2599.

L. Spitzer. *Physics of Fully Ionized Gases*. Interscience Publishers, New York, 2nd edition, 1962.

D. A. Verner and G. J. Ferland. Atomic data for astrophysics. I. radiative recombination rates for H-like, He-like, Li-like, and Na-like ions over a broad range of temperature. *The Astrophysical Journal Supplement Series*, 103:467–473, 1996. doi: 10.1086/192284.

G. S. Voronov. A practical fit formula for ionization rate coefficients of atoms and ions by electron impact: Z=1–28. *Atomic Data and Nuclear Data Tables*, 65(1):1–35, 1997. doi: 10.1006/adnd.1997.0732.